# Resolution Geometry: The Organization of the Ladder

## Dependency Maps, Walls, Gates, Anti-Ladders and Anti-Connectors for L0–L133, Destructor and Dispatcher, and Executable Walks from the Wall-Tap

Matěj Rada
MatejRada@email.cz
Companion to *Resolution Geometry* (print edition)
Licensed under CC BY-NC-ND 4.0

November 11, 2026 (second edition)

**Abstract**

*Resolution Geometry* (RG) states its results as a ladder of 134 rungs, L0–L133, with backward-only dependencies, and 42 companion papers attach to its rungs. Read as a list, this hides its own shape, and it invites two objections: that RG is a scattered toolkit and not one geometry, and that its "One World" reading is philosophy. This companion answers both with things that can be checked.

*Organization.* A deterministic script reads the ladder table and the companion table from the print source, validates them, and regenerates every view: a metro map, hand-overs, walls, gates, longest chains, anti-ladders, audience readings and measurement pipelines. The structure of L0–L123 is unchanged from the first edition; ten rungs are new.

*Three new instruments.*

- An *anti-connector* of two results is their common cone and its maximal elements: where the two results become one structure.
- The *destructor* of a rung or gate is what falls with it: rungs and companions.
- The *dispatcher* returns, for a result and a set of declarations, a typed status and the missing gates.

*Unity audit.* Every one of the 37 non-meta companions rests on the same seven rungs: L0, L1, L2, L6, L9, L18 and L21. No branch hangs from the root alone. The graph stays connected when the root is deleted, and 98.3% of cross-branch rung pairs share more than the root. A scattered-toolkit null model, with every hand-over replaced by a dependency on L0, gives 27 components, 17 root-only branches and 0%.

*Walks.* A dependency table is authored, so the graph alone proves nothing about mathematics. Five executable walks therefore start from one experiment (tap a wall, wait, record) and use one construction at every station: the kernel of the record law, its quotient, the Fisher form on the quotient and its dual. For six root laws the hole dimension is computed by five unrelated methods and every row is constant; a mixed pipeline breaks the row. The walks witness 43 edges of the printed ladder, 22 of them hand-overs, in 12 of 18 branches, each with a destructor.

*One World.* The claim is typed as a gluing statement at L43: one state on a declared cone whose protocol images are the observed laws. A single classical state exists iff the CHSH value is at most 2. One quantum state reaches $2\sqrt{2}$. A PR box would refute the claim. It is a theorem with a test.

Six audit findings are reported for the print. One was applied: the attachment rungs of three recent companions were under-declared.

## Contents

**1 How to use this companion** **3**

**2 Data model** **3**

**3 View 1: the metro map** **4**

**4 View 2: branches and hand-overs** **6**

**5 View 3: walls** **8**

**6 View 4: gates and the assumption budget** **10**

**7 View 5: longest chains** **12**

**8 View 6: anti-ladders** **12**

**9 View 7: audience readings** **15**
9.1 Physics reading: what does the result assume about the world? . . . 16
9.2 Geometry reading: what is forced, what is structure? . . . 16
9.3 Laboratory reading: what can be certified, and how does it fail? . . . 16
9.4 Criticality: what falls if a rung fails . . . 17

**10 View 8: measurement pipelines** **17**

**11 View 9: stations and companions** **18**

**12 View 10: anti-connectors** **20**

**13 View 11: destructor and dispatcher** **22**

**14 Unity audit: one geometry or a scattered toolkit** **25**

**15 Executable walks from the wall-tap** **25**
15.1 One experiment, one construction . . . 25
15.2 Shared fate . . . 26
15.3 The five walks . . . 26
15.4 Witness register . . . 27

**16 One World: a gluing statement with a falsifier** **28**

**17 Objections and answers** **29**

**18 Audit findings for the next print** **29**

**19 Reproducibility** **30**

**A Full register: local versus inherited status** **31**

# 1 How to use this companion

| Reader | Start with | Then |
|---|---|---|
| Physicist | anti-ladders of L109, L105, L90, L80 (§8); assumption matrix (Fig. 3) | gate register (§6); unit, cone and protocol walks (§15) |
| Geometer | unconditional core (Fig. 4); stations (Table 13) | anti-connectors (§12); gauge and wall walks; walls L65, L70, L117–L130 |
| Laboratory / engineer | pipelines (§10); anti-ladder of L113 | dispatcher (§13); wall register; criticality hubs (Table 9) |
| Sceptic | unity audit (§14); shared-fate table (Table 23) | objections (§17); run both scripts and the mutations (§19) |
| Auditor | validation (Table 1); audit findings (§18) | full register (Appendix A) |

Citation remains by rung ("RG, L90"). This companion adds no rung. It reorganizes the statement of record, and where it disagrees with the print it reports an audit finding.

# 2 Data model

**Rungs and local status.** Each rung carries the status printed in the ladder. A rung may carry more than one class (for example "I / N"):

- F, forced;
- I, imported;
- H, conditioned on named hypotheses;
- Γ, conditional on the effective symmetry group;
- N, no-go;
- O, open.

**Effective status (inherited budget).** A rung's *assumption budget* is the union of the hypotheses, and of Γ, over the rung and all its ancestors. A rung whose local status is F but whose budget is non-empty is *forced given* those inherited hypotheses. The register (Appendix A) prints both columns; the difference is the main source of misreading.

**Edges.** An edge $u \to v$ means that $v$ depends on $u$.

- *Logical* edges are the printed dependencies, including short ranges such as L54–L56.
- The *assembly umbrella* L1–L109 of L110 is kept separate (109 edges). It aggregates modules rather than using each one, so it would make every rung an ancestor of the closure layer.

**Branches and stations.** The ladder table has 19 printed section blocks. The 18 branches B0–B17 follow them, with one declared split (quantum, from L38) and two declared merges (symmetry into B0, closure into B13); the script checks this. A branch read as a step on the way from the wall-tap is a *station*. A *hand-over* is a logical edge between branches.

**Walls and gates.** A *wall* is a rung with class N. A *gate* is a rung with class H or Γ; a gate opens the region beyond it under a declaration.

**Anti-ladder.** The anti-ladder of a target $t$ is its ancestor cone, transitively reduced, and drawn with $t$ at the top and the root L0 at the bottom. It answers the question *"what must hold for t?"*. The *forward cone* of a rung is its set of descendants.

**Companions.** A companion paper *attaches* at the rungs listed in the print's companion table. Its *cone* is the union of the anti-ladders of its attachment rungs, and its budget is the budget of that cone. Five companions read every rung (this one among them) and are treated as meta.

**New in this edition.**

- The *anti-connector* of two targets $s, t$ is the common cone $\mathrm{Anc}(s) \cap \mathrm{Anc}(t)$ together with its *meets*, the common ancestors that have no common descendant. A connector (hand-over) shows where one branch feeds another. An anti-connector shows, from the two results downward, where they become one structure.
- The *destructor* of a rung is its forward cone and the set of companions whose cone contains it. The destructor of a gate is what is lost when the gate is not declared.
- The *dispatcher* takes a cone and a set of declared hypotheses. It returns FORCED (empty budget), LOCKED (budget declared) or BLOCKED together with the missing gates.
- The *toolkit null* is the graph in which every hand-over is replaced by a dependency on L0. It is what a collection of unrelated tools sharing only a starting point would look like.
- A *walk* is a chain of rungs from L0 inside the anti-ladder of its endpoint, with executable witnesses.

| Check | Result | Meaning |
|---|---|---|
| O1 | pass | rungs are exactly L0–L133, contiguous |
| O2 | pass | every dependency points backward |
| O3 | pass | the dependency graph is acyclic |
| O4, O5 | pass | L0 is the single root, and every rung is reachable from it |
| O6, O7 | pass | every status parses; every hypothesis is a declared name |
| O8 | pass | nothing depends on an open rung (L132, L133 are leaves) |
| O9 | pass | L0 is the only input |
| O10 | pass | branch boundaries match the printed headers (declared split and merges) |
| O11, O12 | pass | companions attach at existing rungs; every non-meta companion attaches |
| O13 | pass | each of the 16 featured companions is found exactly once |
| U1–U3 | pass | unity checks, and the toolkit null is detected (§14) |
| W-a | pass | the walks battery passed |
| W-b | pass | every witnessed connector is a dependency of the printed ladder |
| W-c | pass | every walk is a chain of ancestors |

Table 1: Validation (script `rg_ladder_org.py`). A failed check aborts the run.

The graph has 134 rungs, 232 logical edges (188 after transitive reduction), 109 umbrella edges and 98 hand-overs between 50 branch pairs. Counting every class a rung carries, the classes are: 53 F, 30 I, 27 H, 30 N, 8 Γ, 2 O and 1 input.

**What changed since the first edition.** Rungs L0–L123 keep their dependencies and statuses. The three open rungs of the first edition were replaced: local classification (L124–L128) and the remainder across walls (L129–L131) are now closed, and two external open problems remain (L132, L133). Hand-overs rose from 89 to 98 and walls from 27 to 30.

## 3 View 1: the metro map

Figure 1 draws every rung once.

- **Lanes** are branches, shaded by audience: blue for geometry, green for physics, orange for laboratory, grey for shared.

- **Horizontal position** is logical depth, the length of the longest chain from L0.
- **Marker shape and colour** give the local class.
- **Grey edges** stay within a branch. **Coloured arcs** are hand-overs, coloured by the exporting branch.
- **The dashed line** marks the assembly umbrella into L110.

Three hubs (L2, L18, L19) feed almost every lane. The physical-distance lane B12 collects hand-overs from seven branches. The chaos lane B11 sits at the far right because its chain is the longest.

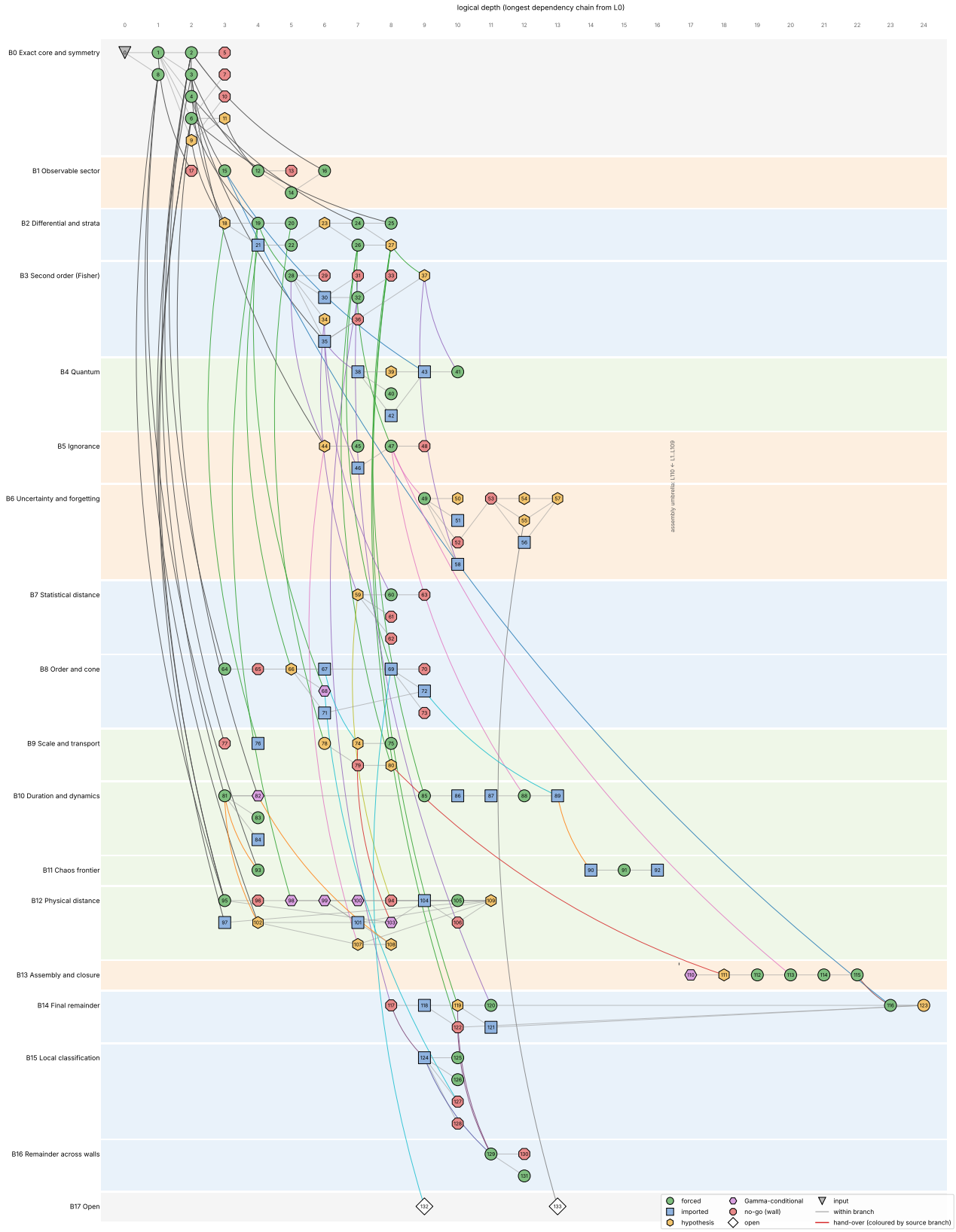



Figure 1: Metro map of L0–L133 (transitively reduced logical edges).

# 4 View 2: branches and hand-overs

Figure 2 collapses each branch to one node. An edge label gives the number of hand-over edges and the first exporting→importing rungs; the complete list is Table 3. Hand-overs are where one branch's

guarantee becomes another branch's premise, so they are where misreadings concentrate.

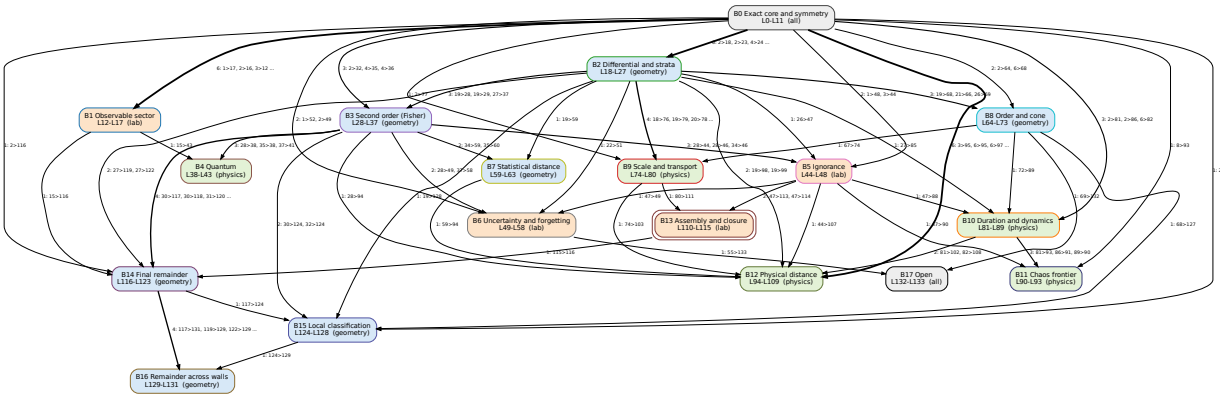



Figure 2: Branch quotient: 18 branches, 50 branch pairs, 98 hand-over edges. B13 receives in addition the assembly umbrella.

| Hand-over | Edges | Exporting rung → importing rung |
|---|---|---|
| B0 → B1 | 6 | L1 (kernel groupoid)→L17, L2 (universal factorization)→L16, L3 (joint kernel = intersection)→L12, L3 (joint kernel = intersection)→L14, L3 (joint kernel = intersection)→L15, L11 (coherent completeness)→L12 |
| B0 → B10 | 3 | L2 (universal factorization)→L81, L2 (universal factorization)→L86, L6 (Gau-Stab-Gamma sequence)→L82 |
| B0 → B11 | 1 | L8 (count rigidity)→L93 |
| B0 → B12 | 6 | L3 (joint kernel = intersection)→L95, L6 (Gau-Stab-Gamma sequence)→L95, L6 (Gau-Stab-Gamma sequence)→L97, L6 (Gau-Stab-Gamma sequence)→L98, L8 (count rigidity)→L95, L8 (count rigidity)→L102 |
| B0 → B14 | 1 | L2 (universal factorization)→L116 |
| B0 → B15 | 1 | L2 (universal factorization)→L126 |
| B0 → B2 | 6 | L2 (universal factorization)→L18, L2 (universal factorization)→L23, L4 (garbling enlarges kernel)→L24, L6 (Gau-Stab-Gamma sequence)→L19, L6 (Gau-Stab-Gamma sequence)→L25, L9 (orbit-quotient comparison)→L18 |
| B0 → B3 | 3 | L2 (universal factorization)→L32, L4 (garbling enlarges kernel)→L35, L4 (garbling enlarges kernel)→L36 |
| B0 → B5 | 2 | L1 (kernel groupoid)→L48, L3 (joint kernel = intersection)→L44 |
| B0 → B6 | 2 | L1 (kernel groupoid)→L52, L2 (universal factorization)→L49 |
| B0 → B8 | 2 | L2 (universal factorization)→L64, L6 (Gau-Stab-Gamma sequence)→L68 |
| B0 → B9 | 1 | L2 (universal factorization)→L77 |
| B1 → B14 | 1 | L15 (coupled refines marginals)→L116 |
| B1 → B4 | 1 | L15 (coupled refines marginals)→L43 |
| B10 → B11 | 3 | L81 (clock groupoid)→L93, L86 (entropy from counts)→L91, L89 (chaos cone fields)→L90 |
| B10 → B12 | 2 | L81 (clock groupoid)→L102, L82 (affine clocks)→L108 |
| B13 → B14 | 1 | L115 (maximal closure)→L116 |
| B14 → B15 | 1 | L117 (coupling complement-relative)→L124 |
| B14 → B16 | 4 | L117 (coupling complement-relative)→L131, L119 (global structure)→L129, L122 (wall monodromy)→L129, L122 (wall monodromy)→L130 |
| B15 → B16 | 1 | L124 (local classifier (word traces))→L129 |
| B2 → B10 | 1 | L27 (stratified quotient category)→L85 |
| B2 → B12 | 2 | L19 (natural descent)→L98, L19 (natural descent)→L99 |
| B2 → B14 | 2 | L27 (stratified quotient category)→L119, L27 (stratified quotient category)→L122 |

| Hand-over | Edges | Exporting rung → importing rung |
|---|---|---|
| B2 → B15 | 1 | L19 (natural descent)→L128 |
| B2 → B3 | 3 | L19 (natural descent)→L28, L19 (natural descent)→L29, L27 (stratified quotient category)→L37 |
| B2 → B5 | 1 | L26 (tangent USC, tame strata)→L47 |
| B2 → B6 | 1 | L22 (one-sided wall directions)→L51 |
| B2 → B7 | 1 | L19 (natural descent)→L59 |
| B2 → B8 | 3 | L19 (natural descent)→L68, L21 (slices and tangent cones)→L66, L26 (tangent USC, tame strata)→L69 |
| B2 → B9 | 4 | L18 (tangent sequence)→L76, L19 (natural descent)→L79, L20 (automatic descent)→L78, L26 (tangent USC, tame strata)→L80 |
| B3 → B12 | 1 | L28 (Fisher natural)→L94 |
| B3 → B14 | 4 | L30 ((R,N,Sigma) reconstruction)→L117, L30 ((R,N,Sigma) reconstruction)→L118, L31 (Fisher does not fix covariance)→L120, L32 (quotient dim p+q+pq)→L117 |
| B3 → B15 | 2 | L30 ((R,N,Sigma) reconstruction)→L124, L32 (quotient dim p+q+pq)→L124 |
| B3 → B4 | 3 | L28 (Fisher natural)→L38, L35 (data processing)→L38, L37 (Hellinger specialization)→L41 |
| B3 → B5 | 3 | L28 (Fisher natural)→L44, L28 (Fisher natural)→L46, L34 (efficient metric)→L46 |
| B3 → B6 | 2 | L28 (Fisher natural)→L49, L37 (Hellinger specialization)→L58 |
| B3 → B7 | 2 | L34 (efficient metric)→L59, L35 (data processing)→L60 |
| B5 → B10 | 1 | L47 (rank walls not certifiable)→L88 |
| B5 → B11 | 1 | L47 (rank walls not certifiable)→L90 |
| B5 → B12 | 1 | L44 (persistent null)→L107 |
| B5 → B13 | 2 | L47 (rank walls not certifiable)→L113, L47 (rank walls not certifiable)→L114 |
| B5 → B6 | 1 | L47 (rank walls not certifiable)→L49 |
| B6 → B17 | 1 | L55 (contraction gate)→L133 |
| B7 → B12 | 1 | L59 (RG distance)→L94 |
| B8 → B10 | 1 | L72 (hyperbolic lock)→L89 |
| B8 → B15 | 1 | L68 (Lorentz forcing)→L127 |
| B8 → B17 | 1 | L69 (tame reduction)→L132 |
| B8 → B9 | 1 | L67 (conformal ray)→L74 |
| B9 → B12 | 1 | L74 (scale cocycle)→L103 |
| B9 → B13 | 1 | L80 (canonical connection)→L111 |

Table 3: Every hand-over edge, listed by branch pair.

## 5 View 3: walls

A wall is a no-go rung: a proved impossibility with a witness. Table 5 lists each wall, its direct successors, and *what opens it*. The walls fall into three kinds:

- *terminal walls*, with no successor, which are permanent boundaries (for example L5, L13, L36, L48);
- *gated walls*, opened by a named hypothesis (for example L53 by H-adm, L65 by H-top, L79 by H-cov);
- *anchored walls*, opened only by counted channels and SI anchors (L94, L96, L63).

A wall is also a destructor in the sense used in the companion derivations: it eliminates an alternative, and what remains is forced. The three new walls are L127, L128 and L130.

| Wall | What it denies | Br. | Direct successors (status) | Opened by |
|---|---|---|---|---|
| L5 | exact layer fixes nothing | B0 | *terminal* | declared higher structure: H-reg (L18), H-mono (L29), H-top (L66), Γ, H-SI |
| L7 | no splitting | B0 | *terminal* | presentation-relative Γ (L6) |
| L10 | rank, separation, completeness distinct | B0 | *terminal* | separate certificate per property (L11, L12) |
| L13 | no task-free selector | B1 | *terminal* | a declared task (L12 then selects exactly) |
| L17 | tolerance not a groupoid | B1 | *terminal* | confidence sets (L16); window walls (L114) |
| L29 | Fisher selected (Cencov) | B3 | *terminal* | H-mono selects Fisher (Čencov) |
| L31 | Fisher does not fix covariance | B3 | L120 (F) | second-order contract (L32, L117); global $[K_N]$ (L120) |
| L33 | shadow is not the law | B3 | *terminal* | protocols on the full law (L15, L43) |
| L36 | kernel order not Blackwell | B3 | *terminal* | none inside the ladder (decision-theoretic order) |
| L48 | law equality not certifiable | B5 | *terminal* | confidence correspondences (L16, L47) |
| L52 | TV rigidity | B6 | L53 (N) | a declared class (L53 to L57) |
| L53 | no uniform forgetting | B6 | L54 (H: adm); L55 (H: adm); L56 (I / N) | H-adm: BV lock (L54), contraction gate (L55) |
| L56 | samples refute, operators lock | B6 | L57 (H: adm) | operator certificates under H-adm (L57) |
| L61 | no angle in rank one | B7 | *terminal* | rank-two strata; anchors |
| L62 | divergences not rulers | B7 | *terminal* | infinitesimal metric (L59); anchors (L102) |
| L63 | experiment-relative distance | B7 | *terminal* | counted channel and SI anchor (L95, L102, L109) |
| L65 | projective typing | B8 | L66 (H: top) | H-top: realization branches (L66) |
| L70 | no finite classification | B8 | *terminal* | tame reduction (L69); enumeration open (L124) |
| L73 | wild class | B8 | *terminal* | window certificates only |
| L77 | entropy selects no connection | B9 | *terminal* | declared covariance (L80) |
| L79 | no law-natural connection | B9 | L80 (H: cov; resp) | H-cov: canonical connection (L80) |
| L94 | no internal unit | B12 | L109 (H: reg, nuis, SI, med, val; Γ) | counted channel (L95) and H-SI (L102) |
| L96 | count loss | B12 | L109 (H: reg, nuis, SI, med, val; Γ) | loss-calibrated channel (H-val in L109) |
| L106 | non-stationary no-go | B12 | L109 (H: reg, nuis, SI, med, val; Γ) | stationary slices only (L104); otherwise open |
| L117 | coupling complement-relative | B14 | L118 (I / N); L123 (F (H: split, type)); L124 (I); L131 (F (I: obstruction theory)) | H-split: declared complement |
| L118 | coupling stabilizer | B14 | L119 (H: split, type); L122 (N / F); L123 (F (H: split, type)) | H-type: constant multiplicity (L119) |
| L122 | wall monodromy | B14 | L123 (F (H: split, type)); L129 (F); L130 (N) | hidden-room descent (L129); obstruction tower (L131) |
| L127 | Lorentzian scalars do not separate | B15 | *terminal* | observer-relative classification (L124) |
| L128 | classification selects no dynamics | B15 | *terminal* | a declared dynamics (duration lane, L81 onward) |
| L130 | no constructible exit paths | B16 | *terminal* | the obstruction tower (L131) replaces a constructible classification |

Table 5: Wall register.

## 6 View 4: gates and the assumption budget

Table 6 lists each hypothesis, the rungs that state it, and the size of the logical region it gates. Figure 3 gives, for 24 audience targets, every hypothesis the target needs: red where the target itself states it, amber where it is inherited.

| Hypothesis | Gated rungs | Downstream |
|---|---|---|
| H-compl | L9 | 109 |
| H-sound | L9 | 109 |
| H-reg | L18, L59, L109 | 100 |
| H-unif | L23, L27, L37 | 45 |
| H-nuis | L34, L59, L109 | 29 |
| H-top | L66 | 20 |
| $\Gamma$ | L68, L82, L98, L99, L100, L103, L109, L110 | 13 |
| H-resp | L80 | 7 |
| H-SI | L102, L109 | 6 |
| H-cov | L80, L111 | 6 |
| H-int | L74 | 6 |
| H-split | L119, L123 | 5 |
| H-type | L119, L123 | 5 |
| H-coh | L11 | 4 |
| H-sens | L44 | 4 |
| H-hell | L37 | 2 |
| H-adm | L54, L55, L57 | 1 |
| H-band | L121 | 1 |
| H-med | L107, L109 | 1 |
| H-task | L39 | 1 |
| H-exch | L50 | 0 |
| H-mono | L29 | 0 |
| H-stat | L108 | 0 |
| H-val | L109 | 0 |

Table 6: Gate register, ordered by the size of the gated region.

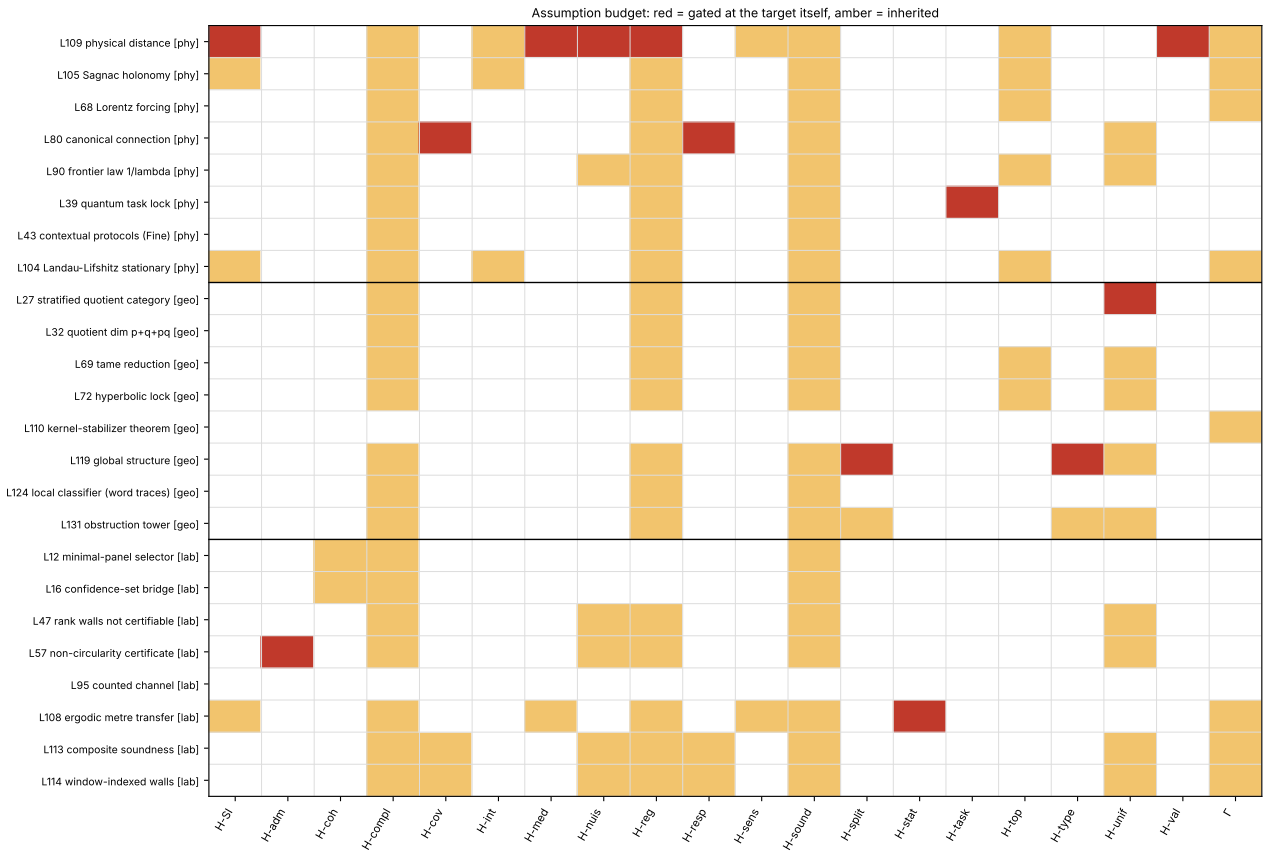



Figure 3: Assumption matrix for physics, geometry and laboratory targets.

**The unconditional core.** 21 rungs have an empty budget (Figure 4). They are L0–L8, L15, L17, L64, L65, L77, L81, L83, L84, L93 and L95–L97. This is what RG asserts with no declaration beyond the observation system itself:

- the kernel and quotient;
- the symmetry sequence;
- coupling and tolerance;
- reachability;
- the clock groupoid, path independence and distortion classes;
- the counted channel and the ratio scale.




Figure 4: The unconditional core (empty assumption budget).

# 7 View 5: longest chains

| Target | Length | Longest logical dependency chain |
|---|---|---|
| L109 | 11 | L0→L1→L2→L18→L21→L66→L67→L74→L103→L104→L105→L109 |
| L113 | 11 | L0→L1→L2→L18→L21→L22→L23→L26→L80→L111→L112→L113 |
| L119 | 10 | L0→L1→L2→L18→L19→L28→L30→L32→L117→L118→L119 |
| L90 | 14 | L0→L1→L2→L18→L21→L22→L23→L24→L27→L85→L86→L87→L88→L89→L90 |
| L80 | 8 | L0→L1→L2→L18→L21→L22→L23→L26→L80 |
| L39 | 8 | L0→L1→L2→L18→L19→L28→L35→L38→L39 |
| L68 | 6 | L0→L1→L2→L18→L21→L66→L68 |
| L131 | 12 | L0→L1→L2→L18→L19→L28→L30→L32→L117→L118→L119→L129→L131 |
| global | 16 | L0→L1→L2→L18→L21→L22→L23→L24→L27→L85→L86→L87→L88→L89→ L90→L91→L92 |

Table 8: Longest logical dependency chains.

The longest ladder is the dynamics–chaos chain: 16 steps to L92, through the strata (L18–L27) and the entropy and Pesin locks. Physical distance needs only 11 steps but has a wide cone (34 ancestors from nine branches). The second-order remainder is reached in 10 steps (L119) and the obstruction tower in 12 (L131). Through the assembly umbrella the deepest rung is L123, at 24 steps.

# 8 View 6: anti-ladders

Each figure answers *"what must hold for this target?"* The target is at the top (double border) and L0 at the bottom; red edges are hand-overs. Bracketed labels name hypotheses stated at that rung. These cones are unchanged from the first edition.

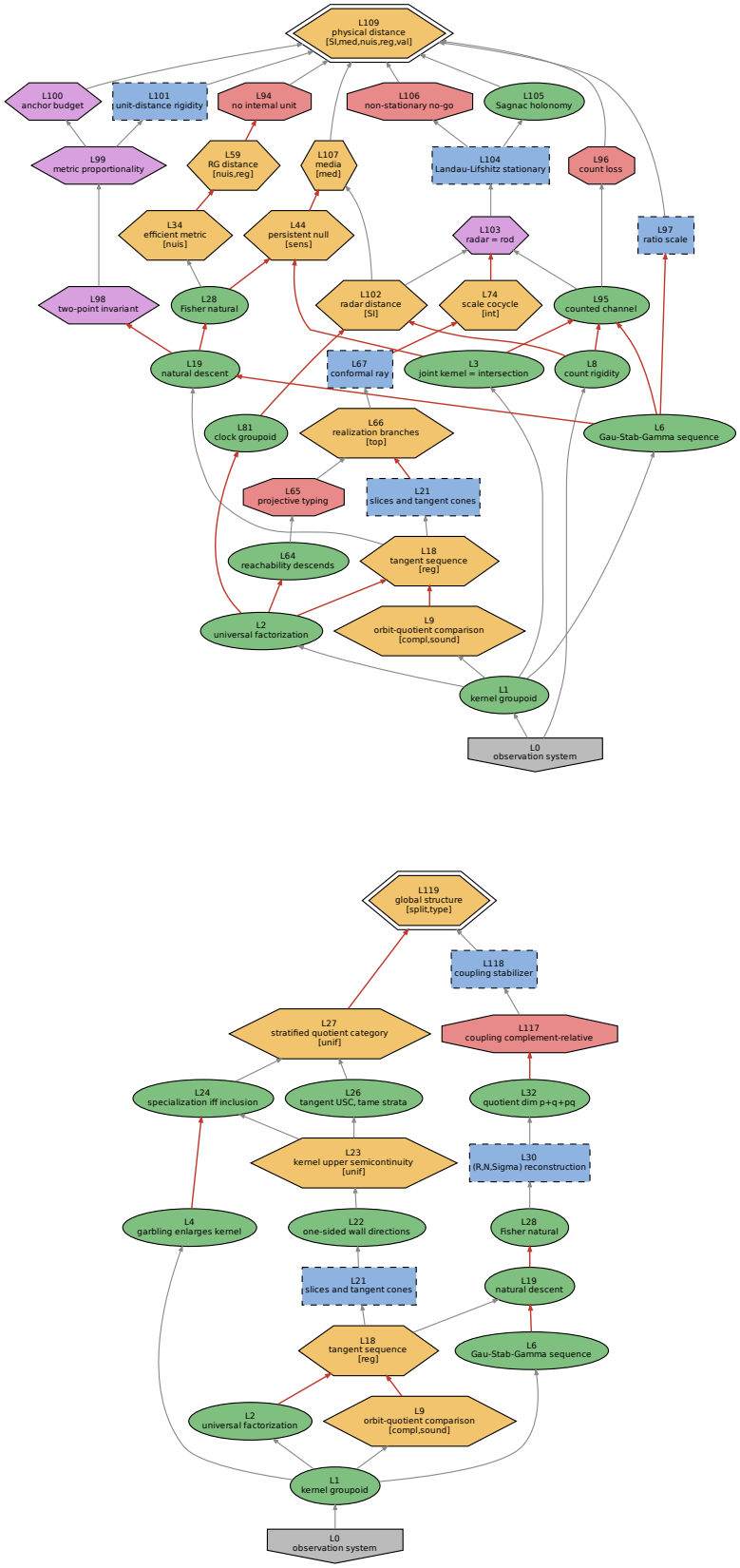



Figure 5: Anti-ladders of L109, physical distance (physics), and L119, the global structure of the second-order remainder (geometry).

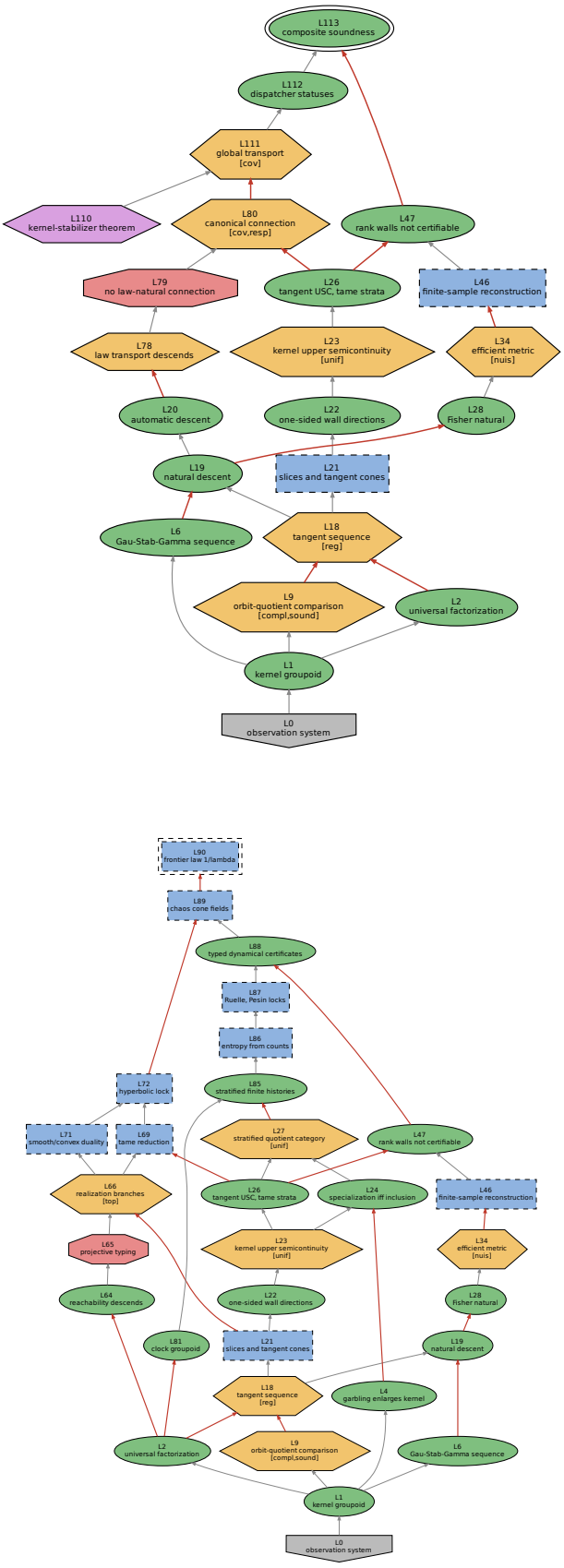



Figure 6: Anti-ladders of L113, composite soundness (laboratory), and L90, the chaos frontier law (physics; the longest chain).

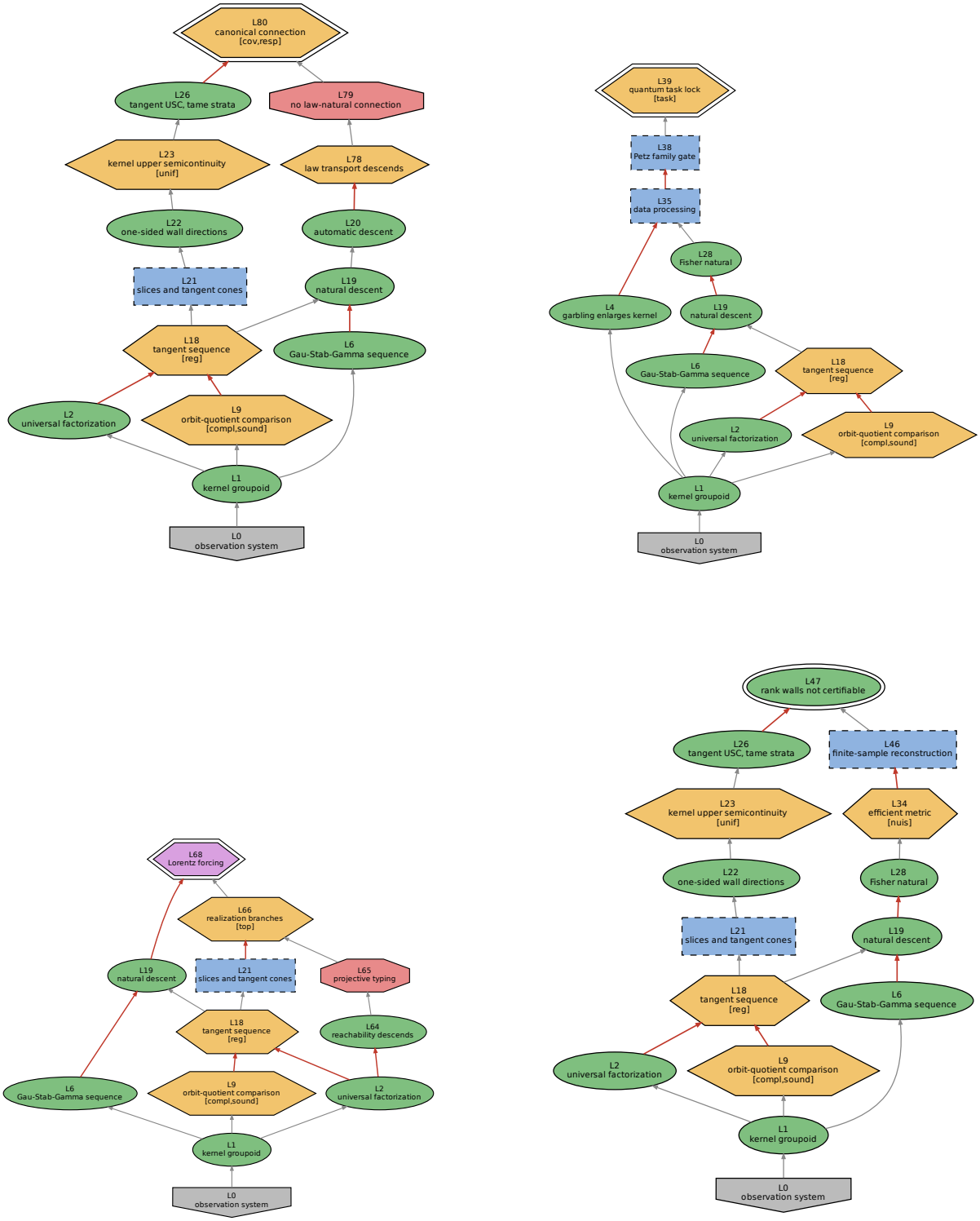



Figure 7: Anti-ladders of L80 (canonical connection), L39 (quantum task lock), L68 (Lorentz forcing) and L47 (rank walls are not certifiable).

## 9 View 7: audience readings

The same ladder supports three readings. They differ in what is taken as given and in what counts as an output.

### 9.1 Physics reading: what does the result assume about the world?

Physics enters only through gates: H-SI, H-med, H-val, H-stat, H-resp, H-cov, H-int, Γ (isotropy, Euclidean motions, time flow), and imports such as Landau–Lifshitz and Pesin. Every output in the duration, chaos and physical-distance lanes (B10–B12) is reached through hand-overs from the core (B0) and the strata (B2).

| Target | Name | Anc. | Chain | Hypothesis budget | Imp. |
|---|---|---|---|---|---|
| L109 | physical distance | 34 | 11 | SI, compl, int, med, nuis, reg, sens, sound, top, val, Γ | 5 |
| L105 | Sagnac holonomy | 19 | 10 | SI, compl, int, reg, sound, top, Γ | 3 |
| L68 | Lorentz forcing | 11 | 6 | compl, reg, sound, top, Γ | 1 |
| L80 | canonical connection | 14 | 8 | compl, cov, reg, resp, sound, unif | 2 |
| L90 | frontier law 1/lambda | 30 | 14 | compl, nuis, reg, sound, top, unif | 10 |
| L39 | quantum task lock | 11 | 8 | compl, reg, sound, task | 2 |
| L43 | contextual protocols (Fine) | 14 | 9 | compl, reg, sound | 4 |
| L104 | Landau-Lifshitz stationary | 18 | 9 | SI, compl, int, reg, sound, top, Γ | 3 |

### 9.2 Geometry reading: what is forced, what is structure?

The geometer reads the F/I spine, the unconditional core, the strata (B2), the cone (B8), the remainder (B14) and its classification (B15, B16). Walls such as L65, L70, L117, L118, L122 and L130 mark where classification stops being unique or constructible.

| Target | Name | Anc. | Chain | Hypothesis budget | Imp. |
|---|---|---|---|---|---|
| L27 | stratified quotient category | 11 | 8 | compl, reg, sound, unif | 2 |
| L32 | quotient dim p+q+pq | 9 | 7 | compl, reg, sound | 1 |
| L69 | tame reduction | 12 | 8 | compl, reg, sound, top, unif | 3 |
| L72 | hyperbolic lock | 14 | 9 | compl, reg, sound, top, unif | 5 |
| L110 | kernel-stabilizer theorem | 0 | 0 | , Γ | 0 |
| L119 | global structure | 19 | 10 | compl, reg, sound, split, type, unif | 4 |
| L124 | local classifier (word traces) | 11 | 9 | compl, reg, sound | 2 |
| L131 | obstruction tower | 23 | 12 | compl, reg, sound, split, type, unif | 6 |

### 9.3 Laboratory reading: what can be certified, and how does it fail?

The laboratory reads certificates and statuses: L12 (instruments), L16 (confidence sets), L47 (walls), L57 (non-circularity), L113 and L114 (soundness, critical margins). Every failure is typed by the dispatcher (L112).

| Target | Name | Anc. | Chain | Hypothesis budget | Imp. |
|---|---|---|---|---|---|
| L12 | minimal-panel selector | 6 | 4 | coh, compl, sound | 0 |
| L16 | confidence-set bridge | 9 | 6 | coh, compl, sound | 0 |
| L47 | rank walls not certifiable | 14 | 8 | compl, nuis, reg, sound, unif | 3 |
| L57 | non-circularity certificate | 21 | 13 | adm, compl, nuis, reg, sound, unif | 5 |
| L95 | counted channel | 5 | 3 | — | 0 |
| L108 | ergodic metre transfer | 17 | 8 | SI, compl, med, reg, sens, sound, stat, Γ | 0 |
| L113 | composite soundness | 22 | 11 | compl, cov, nuis, reg, resp, sound, unif, Γ | 3 |
| L114 | window-indexed walls | 23 | 12 | compl, cov, nuis, reg, resp, sound, unif, Γ | 3 |

### 9.4 Criticality: what falls if a rung fails

| Rung | Name | Forward cone | Branches reached |
|---|---|---|---|
| L0 | observation system | 132 | 18 |
| L1 | kernel groupoid | 130 | 18 |
| L2 | universal factorization | 115 | 18 |
| L9 | orbit-quotient comparison | 109 | 18 |
| L18 | tangent sequence | 102 | 16 |
| L6 | Gau-Stab-Gamma sequence | 92 | 18 |
| L19 | natural descent | 76 | 16 |
| L28 | Fisher natural | 64 | 13 |
| L21 | slices and tangent cones | 61 | 15 |
| L22 | one-sided wall directions | 48 | 13 |
| L23 | kernel upper semicontinuity | 47 | 13 |
| L26 | tangent USC, tame strata | 44 | 13 |
| L4 | garbling enlarges kernel | 31 | 9 |
| L34 | efficient metric | 31 | 9 |
| L46 | finite-sample reconstruction | 23 | 7 |

Table 9: The fifteen rungs with the largest forward cones. L9 ranks fourth, only because of the bundling found in audit A2.

## 10 View 8: measurement pipelines

A pipeline is a concrete task together with the target rungs it must reach. Its budget is the union of the targets' anti-ladders. The column *walls on the way* lists no-go rungs inside the cone. These are conditions the procedure must respect, and they are where an honest report says CORRESPONDENCE or STRUCTURAL WALL instead of LOCKED.

| | Measurement / task | Target rungs | Rungs | Hypothesis budget | Walls on the way |
|---|---|---|---|---|---|
| P1 | Choose sufficient instruments | L12, L14 | 8 | coh, compl, sound | — |
| P2 | Report a value with uncertainty | L16, L49, L51 | 22 | coh, compl, nuis, reg, sound, unif | — |
| P3 | Certify a wall or a rank | L23, L26, L47, L114 | 24 | compl, cov, nuis, reg, resp, sound, unif, Γ | L79 |
| P4 | Calibrated distance (radar, SI) | L102, L103, L107, L109 | 35 | SI, compl, int, med, nuis, reg, sens, sound, top, val, Γ | L65, L94, L96, L106 |
| P5 | Compare and synchronize clocks | L82, L83, L93, L105 | 23 | SI, compl, int, reg, sound, top, Γ | L65 |
| P6 | Forecast horizon of a chaotic system | L88, L90, L91, L92 | 33 | compl, nuis, reg, sound, top, unif | L65 |
| P7 | Non-circular simulation / forgetting | L54, L55, L57 | 22 | adm, compl, nuis, reg, sound, unif | L52, L53, L56 |
| P8 | Quantum estimation or testing | L39, L40, L41 | 21 | compl, hell, reg, sound, task, unif | — |
| P9 | Locomotion / latent transport | L80, L111 | 17 | compl, cov, reg, resp, sound, unif, Γ | L79 |
| P10 | Lock decision for any request | L112, L113, L114, L115 | 25 | compl, cov, nuis, reg, resp, sound, unif, Γ | L79 |
| P11 | Global gluing of local measurements | L119, L121, L129, L131 | 25 | band, compl, reg, sound, split, type, unif | L117, L118, L122 |
| P12 | Classify a second-order datum | L124, L125, L126 | 14 | compl, reg, sound | L117 |

Table 11: Measurement pipelines ("Rungs" is the size of the union cone).

Reading example (P4, calibrated distance). The procedure needs 35 rungs from nine branches. It assumes H-reg, H-nuis, H-sens, H-top, H-int, SI, med and val, plus Γ, together with H-sound and H-compl through L9 (audit A2). Its cone contains four walls, L65, L94, L96 and L106. A laboratory that cannot certify the medium (H-med) or the held-out anchor (H-val) must report the dilation fibre, not a metre. The unit walk of §15 computes that fibre.

## 11 View 9: stations and companions

Table 13 reads each branch as a station on the way from the wall-tap, and says what kind of object it is in the single structure. Two readings were settled by companion papers since the first edition:

- *Uncertainty* (B6) is not a geometry of its own. It is the dual of the Fisher form on the quotient (*Ignorance Before Uncertainty*); walk group W0 checks that the dual is chart-free on the quotient and that the pseudo-inverse on parameter space is not.
- *Coherence* (B4) passes the test for a standalone geometry, conditionally on a non-simplicial record cone and a declared frame (*Coherence Geometry*).

| | Station | Rungs | Role in the single structure | Comp. |
|---|---|---|---|---|
| B0 | Exact core and symmetry | L0–L11 | primitive: records, kernel, quotient, symmetry | 14 |
| B1 | Observable sector | L12–L17 | certificates on the quotient | 12 |
| B2 | Differential and strata | L18–L27 | tangent structure of the quotient (H-reg) | 14 |
| B3 | Second order (Fisher) | L28–L37 | the forced geometry (Fisher) | 12 |
| B4 | Quantum | L38–L43 | coherence: standalone geometry, conditional (non-simplicial cone, declared frame) | 13 |
| B5 | Ignorance | L44–L48 | ignorance: the fibre, prior to uncertainty | 3 |
| B6 | Uncertainty and forgetting | L49–L58 | uncertainty: not standalone, the dual of Fisher on the quotient | 11 |
| B7 | Statistical distance | L59–L63 | distance: integrated Fisher, experiment-relative | 0 |
| B8 | Order and cone | L64–L73 | order and cone; sign by declaration | 11 |
| B9 | Scale and transport | L74–L80 | transport: declared covariance | 2 |
| B10 | Duration and dynamics | L81–L89 | clock groupoid on the quotient | 2 |
| B11 | Chaos frontier | L90–L93 | resolution horizon of dynamics | 1 |
| B12 | Physical distance | L94–L109 | unit: anchors close the dilation hole | 12 |
| B13 | Assembly and closure | L110–L115 | dispatcher: typed statuses | 1 |
| B14 | Final remainder | L116–L123 | second-order global structure | 9 |
| B15 | Local classification | L124–L128 | local classification | 2 |
| B16 | Remainder across walls | L129–L131 | structure across walls | 10 |
| B17 | Open | L132–L133 | open problems (external) | 2 |

Table 13: Stations. "Comp." counts the non-meta companions that attach inside the station. Table 15 is the companion register, computed from the print's companion table.

| Companion | Attaches at | Cone | Br. | Walls | Hypothesis budget |
|---|---|---|---|---|---|
| Physical Path | L28–L32, L119 | 22 | 4 | 4 | compl, mono, reg, sound, split, type, unif |
| Duality of Gravity | L80, L102–L104 | 27 | 6 | 2 | Γ, SI, compl, cov, int, reg, resp, sound, top, unif |
| Light as Emission | L102–L105 | 20 | 6 | 1 | Γ, SI, compl, int, reg, sound, top |
| Cornering | L13, L43, L49–L57, L94 | 36 | 9 | 5 | adm, coh, compl, exch, nuis, reg, sound, unif |
| Undermining Spacetime | L2, L27, L119 | 20 | 4 | 2 | compl, reg, sound, split, type, unif |
| Both Sides of Linear Gravity | L19, L117–L119 | 20 | 4 | 2 | compl, reg, sound, split, type, unif |
| Full Computation Loop | L12, L16, L47, L124 | 24 | 7 | 1 | coh, compl, nuis, reg, sound, unif |
| Ladder Organization (second edition) | all | all | all | — | (reads every rung) |

| Companion | Attaches at | Cone | Br. | Walls | Hypothesis budget |
|---|---|---|---|---|---|
| Realization Atlas | all | all | all | — | (reads every rung) |
| Information Loss | L0–L16, L129–L131 | 36 | 7 | 8 | coh, compl, reg, sound, split, type, unif |
| Physical Shortcut | L64–L68, L102–L109 | 38 | 9 | 4 | Γ, SI, compl, int, med, nuis, reg, sens, sound, stat, top, val |
| Mesh Spacetime | L111, L119, L131 | 30 | 8 | 4 | Γ, compl, cov, reg, resp, sound, split, type, unif |
| GR Without Spacetime | L2, L19, L117–L119 | 20 | 4 | 2 | compl, reg, sound, split, type, unif |
| Spectral Identifiability | L122, L129–L131 | 25 | 6 | 4 | compl, reg, sound, split, type, unif |
| GR Conditional Naturality | L19, L117–L119 | 20 | 4 | 2 | compl, reg, sound, split, type, unif |
| GR Adversarial Brief | all physics companions | all | all | — | (reads every rung) |
| Composition Rigidity | L2, L12, L38–L43 | 27 | 5 | 0 | coh, compl, hell, reg, sound, task, unif |
| Quantum Mechanics | L35, L38–L43, L53–L55, L128 | 38 | 9 | 4 | adm, compl, hell, nuis, reg, sound, task, unif |
| GR–QM Multi-Scale Protocols | L64–L68, L102–L109, L38–L43 | 53 | 11 | 4 | Γ, SI, compl, hell, int, med, nuis, reg, sens, sound, stat, task, top, unif, val |
| Čencov's Theorem in RG | L28–L29, L35, L38–L41 | 22 | 4 | 1 | compl, hell, mono, reg, sound, task, unif |
| Hessian Theorem | L28–L29, L35, L38–L41 | 22 | 4 | 1 | compl, hell, mono, reg, sound, task, unif |
| Rooting Closure | L53–L55, L129–L133 | 39 | 10 | 7 | adm, compl, nuis, reg, sound, split, top, type, unif |
| Structural Closure | L53–L55, L122, L129–L133 | 39 | 10 | 7 | adm, compl, nuis, reg, sound, split, top, type, unif |
| The Case for Engagement | all | all | all | — | (reads every rung) |
| The Perfect Window | all | all | all | — | (reads every rung) |
| Monotonicity and the Lorentz Sign | L28–L29, L35, L38–L41, L64–L68 | 27 | 5 | 2 | Γ, compl, hell, mono, reg, sound, task, top, unif |
| Operational Agreement | L0–L19, L94–L109, L129–L131 | 63 | 13 | 13 | Γ, SI, coh, compl, int, med, nuis, reg, sens, sound, split, stat, top, type, unif, val |
| Experiments, Simulation and Descent | L0–L19, L28–L35, L53–L55, L64–L68 | 44 | 7 | 11 | Γ, adm, coh, compl, mono, nuis, reg, sound, top, unif |
| Wall-Tap Delay, Structural | L1–L22, L46, L52, L70–L82 | 49 | 9 | 11 | Γ, coh, compl, cov, int, nuis, reg, resp, sound, top, unif |
| RG Topology | L0–L22, L35, L129–L131 | 39 | 7 | 9 | coh, compl, reg, sound, split, type, unif |
| Tides Through the RG Window | L1–L22, L53–L55, L129–L131 | 46 | 9 | 11 | adm, coh, compl, nuis, reg, sound, split, type, unif |
| One World, Two Protocols | L38–L43, L53–L55 | 33 | 7 | 2 | adm, compl, hell, nuis, reg, sound, task, unif |
| A World Without Puzzle Pieces | L0–L22, L53–L55, L129–L131 | 46 | 9 | 11 | adm, coh, compl, nuis, reg, sound, split, type, unif |
| From the Wall-Tap (standalone derivation) | L0–L41, L53–L55, L64–L68, L129–L131 | 62 | 11 | 16 | Γ, adm, coh, compl, hell, mono, nuis, reg, sound, split, task, top, type, unif |
| Ignorance Before Uncertainty | L0–L22, L28–L41, L53–L55 | 48 | 7 | 11 | adm, coh, compl, hell, mono, nuis, reg, sound, task, unif |
| Non-Spacetime Attractors and the Source of Gravity | L64–L68, L94–L109 | 38 | 9 | 4 | Γ, SI, compl, int, med, nuis, reg, sens, sound, stat, top, val |
| Physical Dynamics, Locked | L64–L68, L94–L109 | 38 | 9 | 4 | Γ, SI, compl, int, med, nuis, reg, sens, sound, stat, top, val |
| Tensor Derivation (standalone) | L64–L68, L94–L109 | 38 | 9 | 4 | Γ, SI, compl, int, med, nuis, reg, sens, sound, stat, top, val |
| Coherence Geometry | L1, L27–L29, L35, L38–L41 | 22 | 4 | 1 | compl, hell, mono, reg, sound, task, unif |
| Quantum Gravity (second edition) | L1–L2, L19, L38–L48, L64–L68, L94–L109 | 57 | 11 | 5 | Γ, SI, compl, hell, int, med, nuis, reg, sens, sound, stat, task, top, unif, val |
| Observer Bounds | L28, L34–L35, L38–L39, L64–L68, L81–L109 | 63 | 11 | 4 | Γ, SI, compl, int, med, nuis, reg, sens, sound, stat, task, top, unif, val |
| GR Automation | L28–L35, L104–L106, L117, L120 | 40 | 8 | 7 | Γ, SI, compl, int, mono, nuis, reg, sound, split, top, type, unif |

Table 15: Companion register: attachment rungs, size of the cone, branches in the cone, walls in the cone, and the inherited hypothesis budget.

## 12 View 10: anti-connectors

An anti-connector answers the question a sceptic asks of two results that look unrelated: *do they rest on the same thing, and where?* Figure 8 draws two of them. Violet rungs are common to both cones; the meets have a heavy border.

- The quantum task lock (L39) and physical distance (L109) share 8 rungs and meet at L28: both are readings of the forced Fisher geometry.
- Uncertainty on the quotient (L49) and Lorentz forcing (L68) share 8 rungs and meet at L19 and L21: natural descent and the slices.

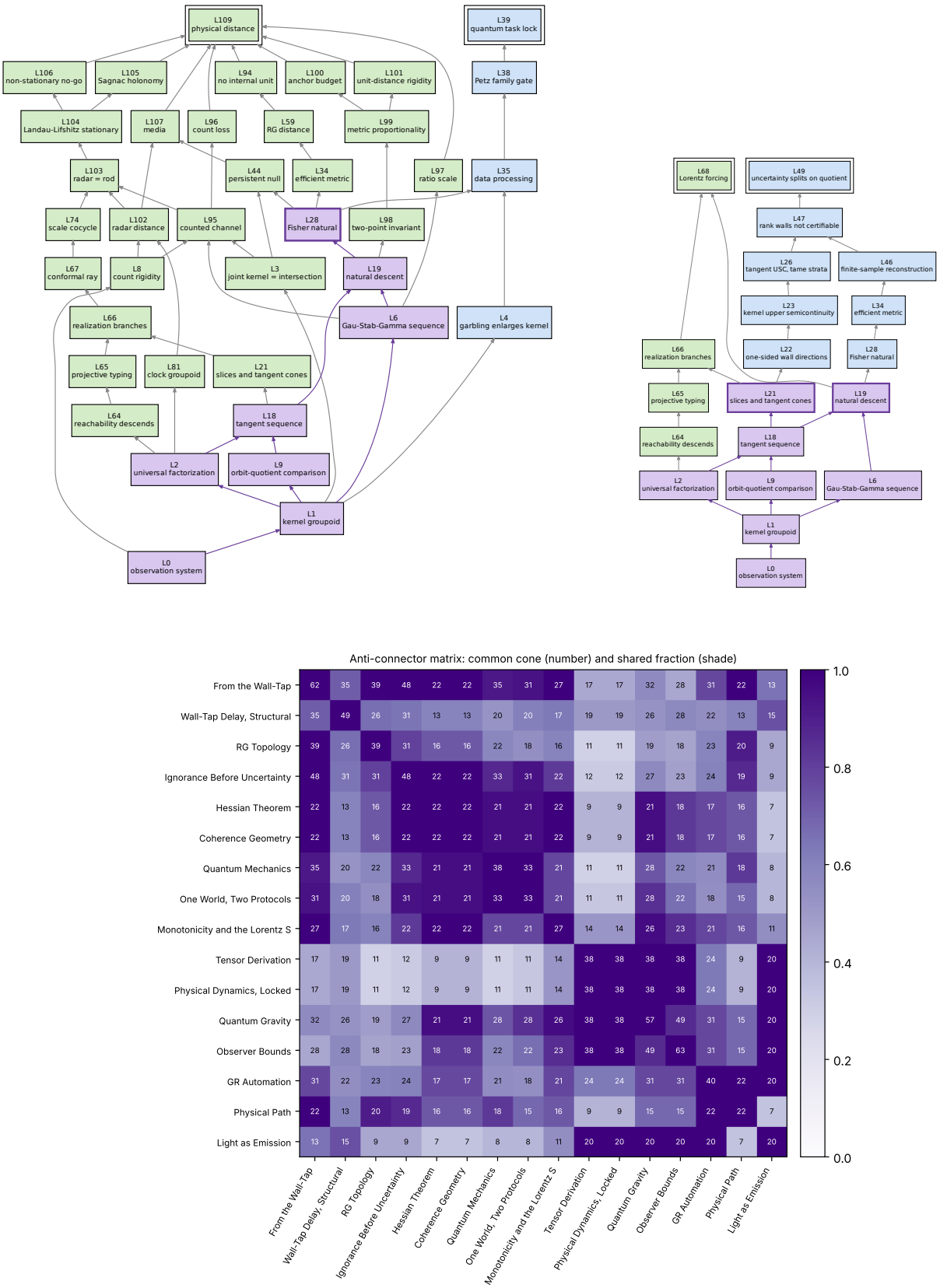



Figure 8: Top: anti-connectors of (L39, L109) and of (L49, L68). Bottom: anti-connector matrix of 16 featured companions. Each cell gives the size of the common cone; the shade is the shared fraction of the smaller cone.

For companions the matrix in Figure 8 gives every pair, and Table 17 names the meets of eight pairs.

| Result A | Result B | Common | Shared | Meets (where the two become one) |
|---|---|---|---|---|
| Ignorance Before Uncertainty | Coherence Geometry | 22 | 1.00 | L29, L40, L41 |
| Coherence Geometry | Tensor Derivation | 9 | 0.41 | L21, L28 |
| Tensor Derivation | Quantum Gravity | 38 | 1.00 | L68, L108, L109 |
| Quantum Gravity | GR Automation | 31 | 0.78 | L27, L34, L35, L105, L106 |
| Observer Bounds | GR Automation | 31 | 0.78 | L27, L34, L35, L105, L106 |
| One World, Two Protocols | Tensor Derivation | 11 | 0.33 | L3, L21, L34 |
| RG Topology | Physical Dynamics, Locked | 11 | 0.29 | L3, L8, L21, L28 |
| Ignorance Before Uncertainty | Observer Bounds | 23 | 0.48 | L3, L8, L27, L39, L47 |

Table 17: Anti-connectors of selected companion pairs.

Three facts stand out.

- *A common floor.* The cones of all 37 non-meta companions contain L0, L1, L2, L6, L9, L18 and L21: the observation system, the kernel groupoid, the universal factorization, the symmetry sequence, the orbit–quotient comparison, the tangent sequence, and slices.
- *No pair is disjoint above the root.* The smallest shared fraction among the featured pairs is 0.29 (*RG Topology* against *Tensor Derivation*, 11 common rungs).
- *Nested cones.* The cone of *Coherence Geometry* lies inside that of *Ignorance Before Uncertainty*, and the cone of *Tensor Derivation* inside that of *Quantum Gravity*. The later papers do not open new foundations.

## 13 View 11: destructor and dispatcher

**Destructor of a gate.** Table 18 lists, for each hypothesis, the rungs lost and the companions affected when it is not declared.

| Gate not declared | Rungs lost | Companions affected |
|---|---|---|
| H-compl | 110 | 37 |
| H-sound | 110 | 37 |
| H-reg | 103 | 37 |
| H-unif | 48 | 32 |
| H-nuis | 32 | 21 |
| H-top | 21 | 17 |
| Γ | 21 | 16 |
| H-split | 7 | 16 |
| H-type | 7 | 16 |
| H-int | 7 | 12 |
| H-coh | 5 | 12 |
| H-task | 2 | 12 |
| H-SI | 8 | 11 |
| H-hell | 3 | 11 |
| H-adm | 4 | 10 |
| H-mono | 1 | 9 |
| H-sens | 5 | 8 |
| H-med | 3 | 8 |
| H-stat | 1 | 8 |
| H-val | 1 | 8 |
| H-cov | 8 | 3 |
| H-resp | 8 | 3 |
| H-exch | 1 | 1 |
| H-band | 2 | 0 |

Table 18: Destructor of each gate.

**Destructor of a rung.** Table 19 lists the fourteen rungs on which the most companions rest.

| Rung | Name | Forward cone | Companions resting on it |
|---|---|---|---|
| L0 | observation system | 132 | 37 |
| L1 | kernel groupoid | 130 | 37 |
| L2 | universal factorization | 115 | 37 |
| L9 | orbit-quotient comparison | 109 | 37 |
| L18 | tangent sequence | 102 | 37 |
| L6 | Gau-Stab-Gamma sequence | 92 | 37 |
| L21 | slices and tangent cones | 61 | 37 |
| L19 | natural descent | 76 | 36 |
| L28 | Fisher natural | 64 | 35 |
| L22 | one-sided wall directions | 48 | 32 |
| L23 | kernel upper semicontinuity | 47 | 32 |
| L26 | tangent USC, tame strata | 44 | 32 |
| L4 | garbling enlarges kernel | 31 | 30 |
| L24 | specialization iff inclusion | 21 | 27 |

Table 19: Destructor of a rung: its forward cone and the companions whose cone contains it.

Two readings. First, H-sound and H-compl appear to gate all 37 companions. This is the bundling of audit A2: with the proposed split of L18, only 12 companions would still inherit them. Second, the rungs on which the most companions rest are exactly the common floor of §12, followed by natural descent (L19) and the Fisher form (L28). A toolkit has no such rungs.

**Dispatcher.** Table 21 runs the dispatcher on the featured companions under two declaration profiles. The *laboratory* profile declares regularity and certification hypotheses (reg, nuis, unif, sens, sound, compl, coh, task, exch, adm). The *physics* profile adds top, int, SI, med, val, stat, mono, hell and Γ. No featured companion is FORCED: each inherits at least reg, sound and compl.

| Companion | Budget | Laboratory profile: missing | Physics profile: missing |
|---|---|---|---|
| From the Wall-Tap | 14 | Γ, hell, mono, split, top, type | split, type |
| Wall-Tap Delay, Structural | 11 | Γ, cov, int, resp, top | cov, resp |
| RG Topology | 7 | split, type | split, type |
| Ignorance Before Uncertainty | 10 | hell, mono | LOCKED |
| Hessian Theorem | 7 | hell, mono | LOCKED |
| Coherence Geometry | 7 | hell, mono | LOCKED |
| Quantum Mechanics | 8 | hell | LOCKED |
| One World, Two Protocols | 8 | hell | LOCKED |
| Monotonicity and the Lorentz Sign | 9 | Γ, hell, mono, top | LOCKED |
| Tensor Derivation | 12 | Γ, SI, int, med, stat, top, val | LOCKED |
| Physical Dynamics, Locked | 12 | Γ, SI, int, med, stat, top, val | LOCKED |
| Quantum Gravity | 15 | Γ, SI, hell, int, med, stat, top, val | LOCKED |
| Observer Bounds | 14 | Γ, SI, int, med, stat, top, val | LOCKED |
| GR Automation | 12 | Γ, SI, int, mono, split, top, type | split, type |
| Physical Path | 7 | mono, split, type | split, type |
| Light as Emission | 7 | Γ, SI, int, top | LOCKED |

Table 21: Dispatcher. "Budget" is the number of gates inherited. The other columns list the gates still missing under each profile; LOCKED means none.

The second-order declarations split and type remain missing for the companions that reach the remainder (L119). For *GR Automation* this is informative: that paper shows that the energy form of the metric supplies the complement which H-split otherwise declares.

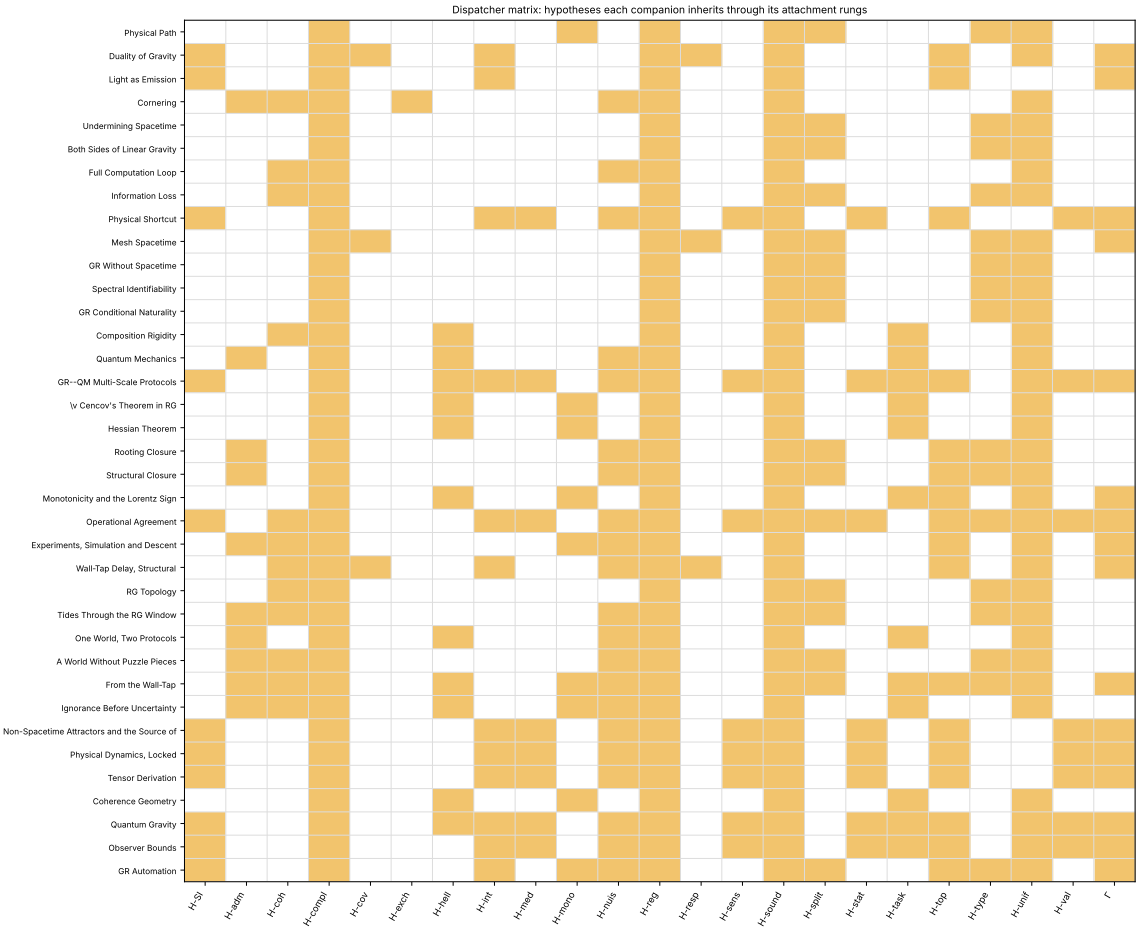



Figure 9: Dispatcher matrix: the hypotheses each companion inherits through its attachment rungs.

# 14 Unity audit: one geometry or a scattered toolkit

The objection is that RG is a set of tools that share a vocabulary and a starting point. In graph terms a toolkit is a star: branches that depend on the root and on nothing else. Table 22 compares the ladder with that null model.

| Quantity | Ladder | Toolkit null |
|---|---|---|
| common spine of all terminal rungs | L0, L1 | L0 |
| terminal rungs (sinks) | 42 | 59 |
| connected components after deleting L0 | 1 | 27 |
| connected components after deleting the common spine | 2 | 27 |
| branches whose only outside dependency is L0 | 0 | 17 |
| smallest common cone of two branch representatives | 4 | 1 |
| median common cone of two branch representatives | 11 | 1 |
| median number of branches inside a rung's cone | 4 | 2 |
| hand-over edges | 98 | 75 |
| cross-branch rung pairs sharing more than L0 | 98.3% | 0.0% |

Table 22: Unity audit. The toolkit null keeps every dependency inside a branch and replaces every hand-over by a dependency on L0.

- The common spine of all 42 terminal rungs is L0 and L1: everything passes through the kernel groupoid. Deleting both isolates only L17, a terminal no-go that depends on L1 alone.
- No branch hangs from the root alone, and the graph remains connected without the root.
- Two branch representatives share at least 4 rungs, 11 in the median.
- The checks U1–U3 require the ladder to pass and the null to fail; a run in which the null is not detected aborts.

**What this does and does not show.** It shows that the printed dependencies are not those of a toolkit. It does not show that the dependencies are right. A dependency column is written by the author, and a hostile reader may treat it as assertion. The next section therefore replaces assertion by computation on one system.

# 15 Executable walks from the wall-tap

## 15.1 One experiment, one construction

The experiment is the one from which RG is derived in *From the Wall-Tap*: tap, wait, record the echo. The parameters are the distance $x$ to the wall and the speed $c$ of the medium; the record is the delay $\tau = 2x/c$ with timing noise. Variants add walls, a gain and a reflectivity, an anchor path of known length, or an attenuation record.
The construction is a single function, `resolve`. From a linearized record law $J$ and a noise covariance it returns:

- the *hole*, the kernel of $J$;
- the resolved complement and the Fisher form $J^\top \Sigma^{-1} J$;
- the *dual*, the efficient covariance, which is defined on the quotient only.

A function *descends* iff its gradient annihilates the hole. Every station below calls this function and no other. For the tap the hole is the dilation generator $(x, c)$; the delay descends, with variance $\sigma^2$ in every chart; the distance does not descend and has no bound (group W0).

## 15.2 Shared fate

If RG were a toolkit, each station would compute its own unrelated quantity. Table 23 gives the hole dimension of six root laws, computed by five methods that share no code beyond the law itself:

**M1** exact rank of the symbolic Jacobian at a rational point (the quotient, L2);

**M2** the dimension of the algebra of scaling fields that annihilate the law identically (symmetry, L6);

**M3** numerical continuation: the number of independent finite displacements at identical law (ignorance, L44);

**M4** the number of vanishing eigenvalues of the Fisher form built from scores (L28);

**M5** the dimension of the set of maximum-likelihood estimates of one seeded data set (uncertainty, L49).

| Root law of the tap | M1 | M2 | M3 | M4 | M5 | Expected |
|---|---|---|---|---|---|---|
| single wall | 1 | 1 | 1 | 1 | 1 | 1 |
| two walls | 1 | 1 | 1 | 1 | 1 | 1 |
| gain contrast | 2 | 2 | 2 | 2 | 2 | 2 |
| anchored | 0 | 0 | 0 | 0 | 0 | 0 |
| anchored calibrated | 0 | 0 | 0 | 0 | 0 | 0 |
| attenuation | 0 | 0 | 0 | 0 | 0 | 0 |
| mixed pipeline (canary) | 1 | 1 | 1 | 0 | 1 | not constant |

Table 23: Shared-fate table (group W1; canary from W10). Every row is constant. In the canary one station is fed a different root law, and the row breaks.

The symmetry orbits are exactly the Fisher radical (subspace equality to $1.5 \times 10^{-8}$). Moving along the hole changes $x$ by 0.44 at identical law, so any estimate of $x$ has an error floor of 0.22 at every sample size. Breaking the construction itself (a mutation of `resolve`) fails eight groups at once.

## 15.3 The five walks

Figure 10 and Table 25 give the walks. Each is a chain of ancestors in the printed ladder (check W-c), and each station is witnessed on the tap.

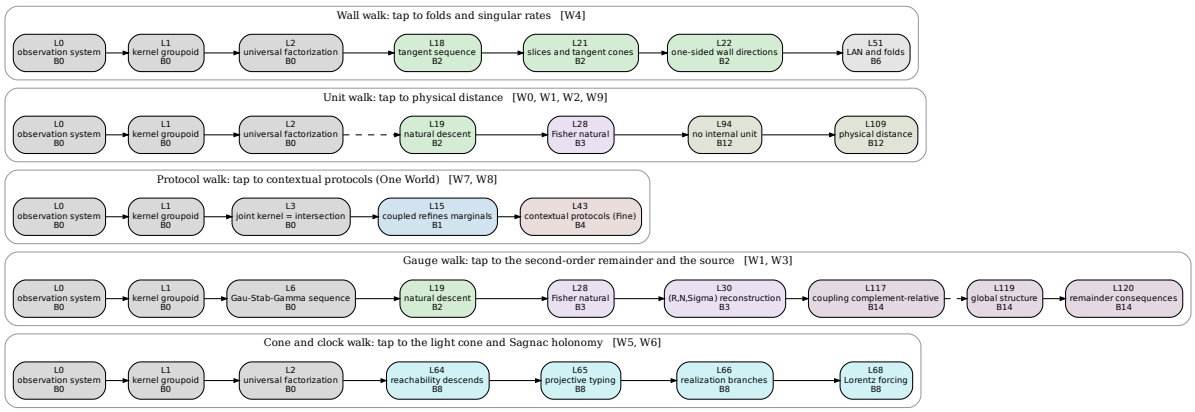



Figure 10: The five walks. Solid arrows are direct dependencies of the printed ladder, dashed arrows pass through an intermediate rung. Brackets name the witness groups.

| Walk | Stations | Witness | Cone | Budget of the endpoint |
|---|---|---|---|---|
| Cone and clock walk | L0→L1→L2→L64→L65→L66→L68 | W5, W6 | 12 | Γ, compl, reg, sound, top |
| Gauge walk | L0→L1→L6→L19→L28→L30→L117→L119→L120 | W1, W3 | 22 | compl, reg, sound, split, type, unif |
| Protocol walk | L0→L1→L3→L15→L43 | W7, W8 | 15 | compl, reg, sound |
| Unit walk | L0→L1→L2→L19→L28→L94→L109 | W0, W1, W2, W9 | 35 | Γ, SI, compl, int, med, nuis, reg, sens, sound, top, val |
| Wall walk | L0→L1→L2→L18→L21→L22→L51 | W4 | 17 | compl, nuis, reg, sound, unif |

Table 25: Walks from the wall-tap: stations, witness groups, cone of the endpoint and its budget.

**Unit walk (W0, W1, W2, W9).** Any number of walls in one medium leaves exactly one hole, the unit; the battery checks one to five walls. Records of the same kind never fix it: a second echo used as an "anchor" leaves the hole open (destructor). Ratios of distances descend, a distance does not. A reference path of known length closes the hole, with $\mathrm{Var}(x)/x^2 = \sigma^2/\tau^2 + \sigma^2/t_{\mathrm{ref}}^2$, matched to $10^{-6}$. A root law without the symmetry (an attenuation record) is closed at every station. On the quotient the Cramér–Rao bound is attained (ratio $0.988 \pm 0.009$).

**Gauge walk (W3).** A layered medium of six cells, with widths and slownesses, seen by the echo alone has 11 null directions. They split exactly as $6 + 5$:

- six relabelings of the cells, which no record removes;
- an interior hole of dimension 5, which interior receivers remove.

With interior receivers the Fisher radical equals the span of the relabeling generators (gap $3 \times 10^{-8}$). Rods remove the relabelings. The uniform relabeling is the dilation of the tap. This is the hand-over to the two most recent companions: the relabeling response is the source tensor (*Tensor Derivation*), and the interior hole is the exterior–interior degeneracy of *GR Automation*.

**Wall walk (W4).** A channel symmetric under $b \mapsto 2d - b$ has a fold at $b = d$. The rank drops there, the fibre is the pair $\{b, 2d - b\}$, and the identification rate is $n^{-0.249}$ at the fold against $n^{-0.501}$ away from it (seeded). A signed channel splits the fold.

**Cone and clock walk (W5, W6).** Before the mediated arrival, the hypotheses "wall" and "no wall" have identical record laws and the optimal error is exactly $1/2$; afterwards it falls to $1.3 \times 10^{-4}$. Arrival times among six points determine a planar configuration only up to the unit. The interval matrix of events has signature $(2, 1)$, and reachability computed from arrivals equals the cone of that form. With wind the one-way times are asymmetric, so the quadratic cone needs the isotropy declaration (gate canary). Three observers with affine clocks satisfy the cocycle condition to $10^{-15}$; the description is unique up to a two-parameter gauge, and a counted clock removes the rate. A reflected overlap has a reflection as holonomy (destructor). Uniform wind gives zero synchronization holonomy exactly; rotation gives $4\Omega A/c^2$ at first order.

**Protocol walk (W7, W8).** See §16.

### 15.4 Witness register

| Group | Rungs witnessed | Ladder edges witnessed (bold: hand-over) |
|---|---|---|
| W0 | L0, L1, L2, L19, L28, L49 | L1→L2, **L19→L28**, **L28→L49**, **L2→L49** |
| W1 | L0, L1, L2, L6, L19, L28, L44 | L1→L2, **L6→L19**, **L19→L28**, **L28→L44** |

| Group | Rungs witnessed | Ladder edges witnessed (bold: hand-over) |
|---|---|---|
| W2 | L6, L28, L94, L95, L97, L102, L109 | **L28→L94**, **L6→L97**, L94→L109, L95→L109, L102→L109 |
| W3 | L6, L28, L30, L31, L117, L119, L120 | **L6→L28**, L28→L30, L30→L31, **L30→L117**, L117→L119, **L31→L120** |
| W4 | L21, L22, L23, L49, L51 | L21→L22, **L22→L51**, L49→L51, L22→L23 |
| W5 | L2, L64, L65, L66, L68 | **L2→L64**, L64→L65, L65→L66, L66→L68, **L6→L68** |
| W6 | L2, L8, L81, L82, L83, L93, L104, L105, L119, L121 | **L2→L81**, L81→L83, **L81→L93**, **L8→L93**, L104→L105, L119→L121 |
| W7 | L1, L4, L28, L35, L38, L39 | L1→L4, **L4→L35**, L28→L35, **L35→L38**, L38→L39 |
| W8 | L3, L15, L42, L43 | **L3→L15**, **L15→L43**, L42→L43 |
| W9 | L28, L34, L46, L49 | L28→L34, **L34→L46**, **L28→L49** |
| W10 | L2, L19, L28 | **L2→L19**, **L19→L28** |

Table 27: Witness register. The script checks every listed edge against the printed ladder (W-b). The eleven groups witness 43 distinct ladder edges, 40 of them direct and 22 of them hand-overs, and they touch 44 rungs in 12 of the 18 branches. Not witnessed on the tap: statistical distance (B7), transport (B9), the assembly (B13), local classification and the remainder across walls (B15, B16), and the open rungs (B17).

## 16 One World: a gluing statement with a falsifier

The objection is that "One World" is a philosophical over-reach. The answer is to say what the claim is.

**The claim.** There is one state, on a declared cone, whose images under the protocols are the observed laws. A protocol is a map of that state to records; a coarser protocol is a garbling of a finer one (L4). The claim is a statement about the existence of a global section. It lives at L43 (contextual protocols), with ancestors L3, L15 and L42.

**On the tap (W7).** One entity: the direct and the reflected wave with delay $\tau$.

- The fine protocol (narrow-band intensities) resolves the delay and both amplitudes (rank 3).
- The coarse protocol (band average) is a garbling of the fine one. Its law is the image of the same law (error 0), its Fisher form is smaller in the positive order, and its rank is 1.
- Over a wide band the fringes wash out (residual $7 \times 10^{-4}$ against 0.68 in a narrow band) and intensities add. The classical rule is the fine law descended to the coarse quotient.
- A which-path protocol has the delay in its hole.
- Without mutual coherence the delay is invisible to the fine protocol (destructor).

Nothing here needs two worlds. There is one law and two resolutions.

**The falsifier (W8).** Take four protocols with consistent marginals.

- A single *classical* state exists iff the CHSH value is at most 2 (Fine's theorem). The battery enumerates the 16 deterministic assignments (maximum 2) and solves the linear program: a section exists at $S = 2$ and none exists for the singlet.
- A single *quantum* state reproduces all four laws at $S = 2\sqrt{2}$, on the boundary of the Tsirelson–Landau–Masanes set (criterion value 1.000 against the bound 1).
- A PR box is consistent on every overlap and has no single quantum state (criterion value 2). Observing one would refute the claim.

So "One World" on a simplex is false in nature, and "One World" on the quantum cone is a theorem-shaped claim that experiment could contradict. The companion *One World, Two Protocols* adds the dynamical version on a single spin. What RG contributes is the typing: macroscopic law is the law descended to a coarse quotient, and the obstruction to a common description is a holonomy, the same object as in the clock walk.

## 17 Objections and answers

| Objection | Answer |
|---|---|
| RG is a scattered toolkit. | Every companion rests on the same seven rungs; no branch hangs from the root; the toolkit null is detected by the same metrics (§12, §14). On one system, five unrelated methods agree on the hole of six root laws, and a mixed pipeline is caught (Table 23). |
| The dependency column is the author's assertion. | Correct, and stated. 43 edges, 22 of them hand-overs, are witnessed on the tap with destructors, and every one is checked against the print (W-b). The edges not witnessed are listed. |
| The walks are a toy. | The tap is the system from which RG is derived, and each group witnesses named rungs. The walks do not prove the general theorems; those are in the print. They show that the stations are one construction on a system small enough to audit. |
| Uncertainty and coherence are separate geometries bolted on. | Uncertainty is the dual on the quotient and is not chart-free off it (W0). Coherence is standalone only under two stated conditions; its cone lies inside the cone of the uncertainty paper (Table 17). |
| One World is philosophy. | It is a gluing statement at L43 with a falsifier (§16). |
| The later papers open new foundations. | Their cones are nested in earlier ones, and their meets are rungs of the ladder (Table 17). L123 states that no new primitive appears on constant-type strata. |
| Graph statistics can be gamed by adding edges. | Yes. That is why the unity audit is not the evidence of record; the walks are. Audit A2 goes the other way: it proposes removing a dependency that inflates the budgets. |

## 18 Audit findings for the next print

The organization exposes places where the print carries more conditionality than a result needs, or less information than it should. The effect of each proposed rewiring is computed on the graph.

| ID | Finding and proposal | Affected | Effect |
|---|---|---|---|
| A1 | The dispatcher L112 depends on L111 (global transport), and therefore on L80, which carries H-cov and H-resp. Totality and soundness of the dispatcher should not need latent transport. *Proposal:* L112←L110; L111 becomes a sibling module. | 6 rungs (L112–L116, L123) | removes cov, resp |
| A2 | L18 (tangent sequence) depends on L9 (orbit–quotient comparison, H-sound, H-compl) only for its completeness-defect clause. *Proposal:* split L18; the tangent sequence needs only L2 and H-reg. | 103 rungs; 37 companions | removes sound, compl from 103 rungs; 12 companions would still inherit them |
| A3 | L80 carries H-resp only for its physical corollary. *Proposal:* move H-resp to a corollary rung, leaving L80 with H-cov. | 8 rungs | removes resp |

| ID | Finding and proposal | Affected | Effect |
|---|---|---|---|
| A4 | The attachment rungs of *Quantum Gravity*, *Observer Bounds* and *Coherence Geometry* were under-declared (L1, L38–L39). Their content uses the cone, physical distance, duration and Fisher lanes. *Applied* in the print's companion table. | 3 companions | budgets now include $\Gamma$, SI, top and others, as they should |
| A5 | No companion attaches inside statistical distance (B7), and the tap witnesses none of its edges. *Proposal:* either a witness or a note that B7 is used only through L59 in physical distance. | 5 rungs | — |
| A6 | The print's companion table did not break across pages, so its later rows were not visible. *Applied.* | print | — |

A2 remains the largest. As printed, almost the whole differential, second-order, cone, dynamics and physical-distance tower, and every companion, formally inherits the declared-groupoid hypotheses. The proposal changes no theorem; it changes what the ladder says each theorem assumes. A1–A3 are not applied.

## 19 Reproducibility

Two scripts, both deterministic.

- `rg_walks.py` (numpy, scipy, sympy; seed 20261111) runs the eleven witness groups and writes `rg_walks_results.json`. It exits nonzero on any failure; a crash inside a group is recorded as a failure of that group.
- `rg_ladder_org.py` (NetworkX, Matplotlib, Graphviz) reads the ladder table and the companion table from the print source, records its SHA-256, runs the checks of Table 1, reads the walks results, and regenerates every figure and table in this paper together with `rg_ladder_org.json`.

Short rung names, branch boundaries, station roles, audience tags, curated wall openers, targets, pipelines, featured companions and declaration profiles are declared in the script; everything else is computed.

**Mutations.** Sixteen mutations were run. Thirteen change the walks battery: the construction, a root law, the symmetry count, the relabeling generators, the fold estimator, the sign of the interval, the reflected overlap, the coherence destructor, the PR box, the noise scale, the mixed pipeline, the closed form, and the garbling. Three change the engine: the null model made equal to the ladder, a shifted branch boundary, and a witnessed edge that is not in the ladder. Every mutation is caught.

**Limits.**

- The organization is only as good as the printed dependency column. A missing dependency in the print is invisible here, and an over-stated one appears as inherited conditionality (A1–A3).
- Companion cones depend on the attachment rungs, which are editorial (A4).
- The walks witness 43 of 232 edges on one system. They do not replace the proofs.
- Method M2 counts scaling symmetries only; the root laws were chosen so that their holes are of that type.
- The Sagnac holonomy is checked at first order. The One World falsifier uses unbiased two-outcome protocols.
- Transitive reduction changes the drawings, not the analysis. Audience tags and station roles are editorial.

# A Full register: local versus inherited status

"Anc." is the number of logical ancestors; "Cone" is the forward cone. Rungs downstream of L110 also inherit, through the umbrella, the hypotheses of whichever modules a request uses.

| Rung | Name | Br. | Local status | Inherited hypotheses | Anc. | Cone |
|---|---|---|---|---|---|---|
| L0 | observation system | B0 | input | — | 0 | 132 |
| L1 | kernel groupoid | B0 | F | — | 1 | 130 |
| L2 | universal factorization | B0 | F | — | 2 | 115 |
| L3 | joint kernel = intersection | B0 | F | — | 2 | 19 |
| L4 | garbling enlarges kernel | B0 | F | — | 2 | 31 |
| L5 | exact layer fixes nothing | B0 | N | — | 3 | 0 |
| L6 | Gau-Stab-Gamma sequence | B0 | F | — | 2 | 92 |
| L7 | no splitting | B0 | N | — | 3 | 0 |
| L8 | count rigidity | B0 | F | — | 1 | 11 |
| L9 | orbit-quotient comparison | B0 | H: sound, compl | compl, sound | 2 | 109 |
| L10 | rank, separation, completeness distinct | B0 | N | compl, sound | 4 | 0 |
| L11 | coherent completeness | B0 | H: coh | coh, compl, sound | 4 | 4 |
| L12 | minimal-panel selector | B1 | F | coh, compl, sound | 6 | 3 |
| L13 | no task-free selector | B1 | N | coh, compl, sound | 7 | 0 |
| L14 | robust observability | B1 | F | coh, compl, sound | 7 | 1 |
| L15 | coupled refines marginals | B1 | F | — | 3 | 3 |
| L16 | confidence-set bridge | B1 | F | coh, compl, sound | 9 | 0 |
| L17 | tolerance not a groupoid | B1 | N | — | 2 | 0 |
| L18 | tangent sequence | B2 | H: reg | compl, reg, sound | 4 | 102 |
| L19 | natural descent | B2 | F | compl, reg, sound | 6 | 76 |
| L20 | automatic descent | B2 | F | compl, reg, sound | 7 | 10 |
| L21 | slices and tangent cones | B2 | I | compl, reg, sound | 5 | 61 |
| L22 | one-sided wall directions | B2 | F | compl, reg, sound | 6 | 48 |
| L23 | kernel upper semicontinuity | B2 | H: unif | compl, reg, sound, unif | 7 | 47 |
| L24 | specialization iff inclusion | B2 | F | compl, reg, sound, unif | 9 | 21 |
| L25 | normalizer descent | B2 | F | compl, reg, sound, unif | 11 | 0 |
| L26 | tangent USC, tame strata | B2 | F / I | compl, reg, sound, unif | 8 | 44 |
| L27 | stratified quotient category | B2 | H: unif | compl, reg, sound, unif | 11 | 19 |
| L28 | Fisher natural | B3 | F | compl, reg, sound | 7 | 64 |
| L29 | Fisher selected (Cencov) | B3 | N / H: mono | compl, mono, reg, sound | 8 | 0 |
| L30 | (R,N,Sigma) reconstruction | B3 | I | compl, reg, sound | 8 | 18 |
| L31 | Fisher does not fix covariance | B3 | N | compl, reg, sound | 9 | 2 |
| L32 | quotient dim p+q+pq | B3 | F | compl, reg, sound | 9 | 16 |
| L33 | shadow is not the law | B3 | N | compl, reg, sound | 10 | 0 |
| L34 | efficient metric | B3 | H: nuis | compl, nuis, reg, sound | 8 | 31 |
| L35 | data processing | B3 | I | compl, reg, sound | 9 | 11 |
| L36 | kernel order not Blackwell | B3 | N | compl, reg, sound | 10 | 0 |
| L37 | Hellinger specialization | B3 | H: unif, hell | compl, hell, reg, sound, unif | 16 | 2 |
| L38 | Petz family gate | B4 | I | compl, reg, sound | 10 | 5 |
| L39 | quantum task lock | B4 | H: task | compl, reg, sound, task | 11 | 1 |
| L40 | Petz identification | B4 | F | compl, reg, sound | 11 | 0 |
| L41 | quantum wall f(0)>0 | B4 | F | compl, hell, reg, sound, task, unif | 19 | 0 |
| L42 | trace rule | B4 | I | compl, reg, sound | 11 | 1 |
| L43 | contextual protocols (Fine) | B4 | I / F | compl, reg, sound | 14 | 0 |
| L44 | persistent null | B5 | H: sens | compl, reg, sens, sound | 9 | 4 |
| L45 | sensor/precision/representation nulls | B5 | F | compl, reg, sens, sound | 10 | 0 |
| L46 | finite-sample reconstruction | B5 | I | compl, nuis, reg, sound | 9 | 23 |
| L47 | rank walls not certifiable | B5 | F | compl, nuis, reg, sound, unif | 14 | 22 |
| L48 | law equality not certifiable | B5 | N | compl, nuis, reg, sound, unif | 15 | 0 |
| L49 | uncertainty splits on quotient | B6 | F / I | compl, nuis, reg, sound, unif | 15 | 10 |
| L50 | de Finetti | B6 | H: exch | compl, exch, nuis, reg, sound, unif | 16 | 0 |
| L51 | LAN and folds | B6 | I / F | compl, nuis, reg, sound, unif | 16 | 0 |
| L52 | TV rigidity | B6 | N | compl, nuis, reg, sound, unif | 16 | 6 |
| L53 | no uniform forgetting | B6 | N | compl, nuis, reg, sound, unif | 17 | 5 |
| L54 | BV mixing lock | B6 | H: adm | adm, compl, nuis, reg, sound, unif | 18 | 1 |
| L55 | contraction gate | B6 | H: adm | adm, compl, nuis, reg, sound, unif | 18 | 2 |
| L56 | samples refute, operators lock | B6 | I / N | compl, nuis, reg, sound, unif | 18 | 1 |

| Rung | Name | Br. | Local status | Inherited hypotheses | Anc. | Cone |
|---|---|---|---|---|---|---|
| L57 | non-circularity certificate | B6 | H: adm | adm, compl, nuis, reg, sound, unif | 21 | 0 |
| L58 | non-dominated strata | B6 | I / F | compl, hell, nuis, reg, sound, unif | 21 | 0 |
| L59 | RG distance | B7 | H: reg, nuis | compl, nuis, reg, sound | 9 | 6 |
| L60 | garbling monotone | B7 | F | compl, nuis, reg, sound | 12 | 1 |
| L61 | no angle in rank one | B7 | N | compl, nuis, reg, sound | 10 | 0 |
| L62 | divergences not rulers | B7 | N | compl, nuis, reg, sound | 10 | 0 |
| L63 | experiment-relative distance | B7 | N | compl, nuis, reg, sound | 13 | 0 |
| L64 | reachability descends | B8 | F | — | 3 | 22 |
| L65 | projective typing | B8 | N | — | 4 | 21 |
| L66 | realization branches | B8 | H: top | compl, reg, sound, top | 8 | 20 |
| L67 | conformal ray | B8 | I | compl, reg, sound, top | 9 | 7 |
| L68 | Lorentz forcing | B8 | Γ | compl, reg, sound, top, Γ | 11 | 1 |
| L69 | tame reduction | B8 | I / F | compl, reg, sound, top, unif | 12 | 8 |
| L70 | no finite classification | B8 | N | compl, reg, sound, top, unif | 13 | 0 |
| L71 | smooth/convex duality | B8 | I | compl, reg, sound, top | 9 | 5 |
| L72 | hyperbolic lock | B8 | I / F | compl, reg, sound, top, unif | 14 | 4 |
| L73 | wild class | B8 | N | compl, reg, sound, top, unif | 13 | 0 |
| L74 | scale cocycle | B9 | H: int | compl, int, reg, sound, top | 10 | 6 |
| L75 | strict transport | B9 | F | compl, int, reg, sound, top | 11 | 0 |
| L76 | induced connection | B9 | I | compl, reg, sound | 5 | 0 |
| L77 | entropy selects no connection | B9 | N | — | 3 | 0 |
| L78 | law transport descends | B9 | F / H | compl, reg, sound | 8 | 9 |
| L79 | no law-natural connection | B9 | N | compl, reg, sound | 9 | 8 |
| L80 | canonical connection | B9 | H: cov; resp | compl, cov, reg, resp, sound, unif | 14 | 7 |
| L81 | clock groupoid | B10 | F | — | 3 | 20 |
| L82 | affine clocks | B10 | Γ | , Γ | 5 | 1 |
| L83 | path independence | B10 | F | — | 4 | 0 |
| L84 | coherence duration (BCH) | B10 | I | — | 4 | 0 |
| L85 | stratified finite histories | B10 | F | compl, reg, sound, unif | 13 | 7 |
| L86 | entropy from counts | B10 | I | compl, reg, sound, unif | 14 | 6 |
| L87 | Ruelle, Pesin locks | B10 | I | compl, reg, sound, unif | 15 | 5 |
| L88 | typed dynamical certificates | B10 | F | compl, nuis, reg, sound, unif | 22 | 4 |
| L89 | chaos cone fields | B10 | I | compl, nuis, reg, sound, top, unif | 29 | 3 |
| L90 | frontier law 1/lambda | B11 | I / F | compl, nuis, reg, sound, top, unif | 30 | 2 |
| L91 | duality 2 lambda, growth h | B11 | F | compl, nuis, reg, sound, top, unif | 31 | 1 |
| L92 | lambda/ln2 bits per step | B11 | I | compl, nuis, reg, sound, top, unif | 32 | 0 |
| L93 | offset/rate/warp | B11 | F | — | 5 | 0 |
| L94 | no internal unit | B12 | N | compl, nuis, reg, sound | 10 | 1 |
| L95 | counted channel | B12 | F | — | 5 | 6 |
| L96 | count loss | B12 | N | — | 6 | 1 |
| L97 | ratio scale | B12 | I | — | 3 | 1 |
| L98 | two-point invariant | B12 | Γ | compl, reg, sound, Γ | 7 | 5 |
| L99 | metric proportionality | B12 | Γ | compl, reg, sound, Γ | 8 | 4 |
| L100 | anchor budget | B12 | Γ | compl, reg, sound, Γ | 9 | 1 |
| L101 | unit-distance rigidity | B12 | I | compl, reg, sound, Γ | 9 | 1 |
| L102 | radar distance | B12 | H: SI | SI | 5 | 7 |
| L103 | radar = rod | B12 | Γ | SI, compl, int, reg, sound, top, Γ | 17 | 4 |
| L104 | Landau-Lifshitz stationary | B12 | I | SI, compl, int, reg, sound, top, Γ | 18 | 3 |
| L105 | Sagnac holonomy | B12 | F | SI, compl, int, reg, sound, top, Γ | 19 | 1 |
| L106 | non-stationary no-go | B12 | N | SI, compl, int, reg, sound, top, Γ | 19 | 1 |
| L107 | media | B12 | H: med | SI, compl, med, reg, sens, sound | 13 | 2 |
| L108 | ergodic metre transfer | B12 | H: stat | SI, compl, med, reg, sens, sound, stat, Γ | 17 | 0 |
| L109 | physical distance | B12 | H: reg, nuis, SI, med, val; Γ | SI, compl, int, med, nuis, reg, sens, sound, top, val, Γ | 34 | 0 |
| L110 | kernel-stabilizer theorem | B13 | Γ | , Γ | 0 | 7 |

| Rung | Name | Br. | Local status | Inherited hypotheses | Anc. | Cone |
|---|---|---|---|---|---|---|
| L111 | global transport | B13 | H: cov | compl, cov, reg, resp, sound, unif, Γ | 16 | 6 |
| L112 | dispatcher statuses | B13 | F | compl, cov, reg, resp, sound, unif, Γ | 17 | 5 |
| L113 | composite soundness | B13 | F | compl, cov, nuis, reg, resp, sound, unif, Γ | 22 | 4 |
| L114 | window-indexed walls | B13 | F | compl, cov, nuis, reg, resp, sound, unif, Γ | 23 | 3 |
| L115 | maximal closure | B13 | F | compl, cov, nuis, reg, resp, sound, unif, Γ | 24 | 2 |
| L116 | task residual | B14 | F | compl, cov, nuis, reg, resp, sound, unif, Γ | 27 | 1 |
| L117 | coupling complement-relative | B14 | N | compl, reg, sound | 10 | 14 |
| L118 | coupling stabilizer | B14 | I / N | compl, reg, sound | 11 | 8 |
| L119 | global structure | B14 | H: split, type | compl, reg, sound, split, type, unif | 19 | 6 |
| L120 | remainder consequences | B14 | F | compl, reg, sound, split, type, unif | 21 | 1 |
| L121 | gluing class H2 | B14 | I / H: band | band, compl, reg, sound, split, type, unif | 20 | 1 |
| L122 | wall monodromy | B14 | N / F | compl, reg, sound, unif | 19 | 4 |
| L123 | no new primitive | B14 | F (H: split, type) | band, compl, cov, nuis, reg, resp, sound, split, type, unif, Γ | 40 | 0 |
| L124 | local classifier (word traces) | B15 | I | compl, reg, sound | 11 | 7 |
| L125 | simple-spectrum sign chart | B15 | F | compl, reg, sound | 12 | 0 |
| L126 | short descriptors lose orbits | B15 | F | compl, reg, sound | 12 | 0 |
| L127 | Lorentzian scalars do not separate | B15 | N | compl, reg, sound, top, Γ | 17 | 0 |
| L128 | classification selects no dynamics | B15 | N | compl, reg, sound | 12 | 0 |
| L129 | hidden-room descent | B16 | F | compl, reg, sound, split, type, unif | 22 | 2 |
| L130 | no constructible exit paths | B16 | N | compl, reg, sound, split, type, unif | 23 | 0 |
| L131 | obstruction tower | B16 | F (I: obstruction theory) | compl, reg, sound, split, type, unif | 23 | 0 |
| L132 | tame type enumeration | B17 | O | compl, reg, sound, top, unif | 13 | 0 |
| L133 | forced turbulence forgetting | B17 | O | adm, compl, nuis, reg, sound, unif | 19 | 0 |